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Deforestation in the Colombian amazon biome: socioeconomic and socio-environmental conflict drivers

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ABSTRACT

Colombia is one of the world's most biodiverse countries, home to over 10% of the planet's biodiversity. Much of this richness is found in the Amazon biome, which spans over 43% of the country's land surface. Despite its ecological importance, the region is subject to growing threats from deforestation, leading to loss of biodiversity and degradation of ecosystem services. Key drivers of this deforestation include socio-environmental conflicts, such as land conversion, coca cultivation, and illegal mining, all of which are compounded by Colombia's longstanding internal armed conflict. Socioeconomic conditions in local communities further reinforce these drivers, yet limited access to reliable data constrains understanding and has led to a gap in literature. This study examines the links between socioeconomic factors, socio-environmental conflicts, and deforestation across the Amazon biome in Colombia. Using different sources of information, we construct a broad set of indicators and use econometric methods and machine learning (ML) techniques to identify the main drivers of forest loss. The results show that while the contribution of coca cultivation to deforestation has declined, land grabbing has emerged as a leading driver. Mining causes significant disturbances to the Amazon biome, but its link to deforestation is marginal. Other consistent predictors include poverty and the presence of basic trails and roads. The analysis also confirms that the northwestern zone remains the most threatened subregion, and that forest conservation is positively associated with the presence of Indigenous communities.

1. Introduction

The Amazon region spans eight South American countries—Bolivia, Brazil, Colombia, Ecuador, Guyana, Peru, Suriname, and Venezuela—as well as one overseas territory (French Guiana). Together they cover a total area of 8,470,209 km^2 (RAISG, 2022). This vast region harbors the world's largest tropical rainforest and is characterized by exceptional endemism: 22% of the planet's species are found exclusively within its boundaries (Miloslavich et al., 2010). The Amazon's vast hydrological network supplies freshwater and nutrients to the oceans, accounting for 15–20% of the world's river discharge. Its vegetation and soils sequester an estimated 150–200 billion tons of carbon annually—equivalent to 15–20 years of global CO_2 emissions (Flores et al., 2024). Preserving this ecosystem is essential not only for the region's approximately 33.5 million inhabitants—63% of whom live in urban areas—but also for protecting over 410 Indigenous groups, who collectively occupy 27.5% of its territory (RAISG, 2020).

In Colombia, the Amazon biome spans 483,164 km^2, accounting for 42.3% of Colombia's national territory and 5.71% of the entire Amazon region (Guio-Rodriguez & Rojas-Suarez, 2019). The Colombian Amazon boasts at least 170 types of ecosystems (Guio-Rodriguez & Rojas-Suarez, 2019). Of the 4932 vertebrate species (fish, birds, amphibians, reptiles, and mammals) recorded in Colombia at the time, 39.8% inhabit this region (Romero et al., 2008). The Colombian Amazon is also home to between 1300 and 1500 bird species, representing 75% of all bird species found in the country (Cárdenas López et al., 2019).

The Colombian Amazon biome (CAB) provides a wide range of ecosystem services at local, regional, and global scales. These include food and fiber production, cultural and intangible services, carbon sequestration, water and climate regulation, recreation and ecotourism, and essential habitat for a vast number of species. The region is also characterized by its rich hydrographic network that supports freshwater

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supply, transport, and climate regulation—not only for local communities but also for populations beyond Colombia's borders. One of its most significant contributions is the redistribution of water across the entire national territory through the phenomenon known as flying rivers (Fundación para la Conservación y el Desarrollo Sostenible (FCDS), 2024).

Despite its critical role in biodiversity conservation, ecosystem services, and the protection of Indigenous cultures, the Colombian Amazon is under mounting pressure. These pressures increasingly threaten its forests, which underpin the complex ecological networks that sustain their exceptional biodiversity.

Understanding the Amazon biome requires not only identifying the dynamics and intensity of its degradation—along with its direct and underlying drivers—but also examining the characteristics of the diverse populations inhabiting the region.

Using data from the 2005 census, Meisel Roca et al. (2013) found that the Colombian Amazon was one of the country's least prosperous regions, with living conditions well below the national average. This was largely due to geographical isolation, limited access to legal economic opportunities, and other structural constraints.

The population of the Amazon both shapes and is shaped by the conservation and management of the biome, giving rise to continuous interactions and feedback loops. Understanding the characteristics of households within the Amazon biome is therefore essential for designing policies that support both forest protection and human development. However, to our knowledge, no recent and comprehensive demographic and socioeconomic analysis of the region has been conducted using data from Colombia's latest national census (2018). While some studies address specific indicators—such as population density and poverty (Mena et al., 2006; Murad & Pearse, 2018)—broader analyses remain limited. By contrast, a substantial body of literature has explored the drivers of deforestation in the Amazon using a range of methodological approaches. These include quantitative statistical models, spatiotemporal analyses, qualitative approaches, and hybrid methods that combine these perspectives (Armenteras et al., 2013a, 2017; Hoffmann et al., 2018; Murillo-Sandoval et al., 2020; González-González et al., 2021a; González-González et al., 2021a). Some studies examine changes over time using time-series data (Murillo-Sandoval et al., 2018; Bautista-Cespedes et al., 2021), while others explore municipal-level differences using cross-sectional data (Meisel Roca et al., 2013). Panel data analyses have also been used to integrate both approaches (Mena et al., 2006; Cantillo & Garza, 2022).

This study contributes by constructing a novel and internally consistent set of indicators for the CAB, spanning four key dimensions: (1) socioeconomic and demographic factors, (2) socio-environmental conflicts, (3) freshwater supply and demand, and (4) terrestrial ecosystems (forests). By systematically organizing and integrating information across these dimensions, the paper provides a territorially coherent representation of the multiple processes shaping deforestation dynamics in a region characterized by severe data scarcity and fragmented information systems. Building on this data framework, the study combines econometric and machine learning approaches to examine interactions among indicators and to assess their relevance in explaining patterns of forest loss.

We examine the correlational structure of deforestation dynamics across the Amazon biome in Colombia, focusing on the interdependence with land grabbing, illicit crop cultivation, and illegal mining. Given the complex dynamics and relationships among variables, rather than pursuing causal identification, we adopt a structural–correlational approach to characterize the interrelated dynamics among key processes highlighted in the literature.

This document is structured as follows. Following this introduction, Section 2 provides a literature review on deforestation drivers on the Amazon. Section 3 outlines the data and methodological approaches used in this study. Section 4 presents the results of the analysis of interactions between indicators and categories through econometric methods to explain deforestation patterns. Finally, Section 5 offers a discussion on the relevance of these findings for global conservation debates.

2. Literature review

Deforestation in the Amazon arises from the interplay of multiple factors involving biophysical, economic, demographic, and institutional factors. The literature typically distinguishes between underlying structural drivers, conflict-related dynamics, direct land-use drivers, and territorial or biophysical conditions that shape the spatial feasibility of land-use change. Studies increasingly emphasize that forest loss cannot be explained by isolated factors, but rather by the interaction between structural conditions, conflict dynamics, and land-use processes that ultimately translate into deforestation. In particular, research highlights the links between armed conflict and illicit crop cultivation (Armenteras et al., 2013b; Quiroga-Angel et al., 2022; Hoffmann et al., 2018; Rincón-Ruiz et al., 2013; Berenguer et al., 2021), as well as the association between coca cultivation and poverty in the Colombian Amazon (Hoffmann et al., 2018; Dávalos et al., 2016; Dávalos & Dávalos, 2020). These interactions are particularly complex in this region due to its history of armed conflict, institutional fragility, and frontier expansion.

Structural drivers include poverty, demographic dynamics, weak governance, and broader institutional conditions (Bautista-Cespedes et al., 2021; Armenteras et al., 2006; Murad & Pearse, 2018). Conflict-related drivers encompass the presence of armed groups, forced displacement, and territorial control (Cantillo & Garza, 2022; Bautista-Cespedes et al., 2021; Liévano-Latorre et al., 2021; Armenteras et al., 2019; Clerici et al., 2020; Murillo-Sandoval et al., 2020; Prem et al., 2020; Fergusson Talero et al., 2014). Direct drivers include land grabbing, agricultural expansion (Busch & Ferretti-Gallon, 2023; Armenteras et al., 2013a, 2017; González-González et al., 2021a; Hänggli et al., 2023; Lim et al., 2017; Tebbutt et al., 2021), illicit crop cultivation (Dávalos & Dávalos, 2021; Dávalos et al., 2016), and mining (González-González et al., 2021b; Hänggli et al., 2023). Finally, territorial and biophysical conditions—such as road infrastructure, accessibility, and landscape characteristics—shape where and how these processes unfold.

An alternative but closely related classification proposed in the literature distinguishes between direct and underlying drivers of deforestation (Bautista-Cespedes et al., 2021). While this framework groups drivers differently, it similarly recognizes that direct land-use activities are shaped by broader socioeconomic and institutional conditions. Under this classification, direct drivers include illicit crop cultivation, agricultural expansion, extensive livestock farming, infrastructure development, and timber extraction. Underlying drivers encompass poverty, armed conflict, forced displacement, land grabbing, and institutional conditions (Armenteras et al., 2006, 2013a, 2019; Etter et al., 2006; Sánchez-Cuervo & Aide, 2013; Castro-Nunez et al., 2017a; Murad & Pearse, 2018; Hoffmann et al., 2018; Landholm et al., 2019; Negret et al., 2019; Furumo & Lambin, 2020).

2.1. Structural socioeconomic conditions

In frontier contexts such as the Colombian Amazon, these structural conditions shape the incentives and constraints faced by different actors involved in land-use decisions. Some studies conceptualize these processes through the distinction between impoverishment and frontier models (Rudel & Roper, 1997). Impoverishment models emphasize the role of poverty and limited livelihood opportunities in driving small-scale agricultural expansion and illicit crop cultivation (Dávalos & Dávalos, 2021). In contrast, frontier models highlight the role of powerful actors and capital-intensive economic activities—including large-scale livestock production and agribusiness—in driving forest conversion through land accumulation and speculative land markets.

Only a limited number of studies include socioeconomic or demographic variables; however, those that do incorporate factors such as

education (Mena et al., 2006), poverty (Mena et al., 2006), population density (Hänggli et al., 2023; Mena et al., 2006; Quiroga-Angel et al., 2022), migration (Dávalos et al., 2016), urbanization, and road infrastructure (Asner et al., 2006; Dávalos et al., 2016; Hänggli et al., 2023; Mena et al., 2006; Merry et al., 2009). Several of these studies also incorporate biophysical variables.

Demographic factors have also been identified as important underlying drivers of deforestation. Population growth, migration flows, and settlement dynamics often accompany the expansion of agricultural frontiers and contribute to increased pressure on forest ecosystems (Dávalos et al., 2016; Hänggli et al., 2023). However, relatively few studies incorporate detailed socioeconomic indicators such as education levels, poverty rates, or access to basic services. Mena et al. (2006), for example, analyze deforestation in the Ecuadorian Amazon using parish-level census data and find that population density, road density, and education levels significantly influence forest loss dynamics. Hänggli et al. (2023) also find that higher levels of education are associated with greater deforestation in both the Ecuadorian and Brazilian Amazon. While these findings provide valuable insights, several authors caution that deforestation drivers differ substantially across Amazonian countries and cannot be directly extrapolated to the Colombian Amazon (Armenteras et al., 2006; Viña & Estévez, 2013).

The lack of recent, municipal-level socioeconomic and demographic data on the population of the Colombian Amazon limits a comprehensive understanding of the biome and poses a significant barrier to the development of sustainable strategies for the region. At the same time, the wide range of studies analyzing deforestation drivers has advanced understanding of forest loss in the Colombian Amazon. However, a comprehensive model that captures the relationships between direct and underlying drivers of deforestation in the CAB—and the interactions among different types of drivers—has yet to be developed or empirically assessed.

2.2. Armed conflict and territorial control

A second strand of literature emphasizes the role of armed conflict and territorial control in shaping deforestation dynamics in the Colombian Amazon. The relationship between conflict and forest loss remains highly debated, with mixed evidence regarding its direction and magnitude.

There is no clear consensus on how armed conflict and deforestation relate. Some studies find that higher conflict intensity exacerbates forest loss (Liévano-Latorre et al., 2021; Negret et al., 2019; Bautista-Cespedes et al., 2021), driven by factors such as forced displacement (Dávalos et al., 2016), land grabbing (Cantillo & Garza, 2022; Negret et al., 2019), and the expansion of illicit crops (González-González et al., 2021b). Other studies suggest that the presence of armed groups can have conservation effects by restricting infrastructure development and limiting productive investments (Murillo-Sandoval et al., 2020). In addition, forced and mass displacement may contribute to forest regeneration in abandoned areas (Hoffmann et al., 2018; Castro-Nunez et al., 2017b; Murillo-Sandoval et al., 2020; Clerici et al., 2020).

Bautista-Cespedes et al. (2021) show that armed conflict has contributed to population displacement and that the presence of anti-personnel mines has, in some cases, promoted forest regeneration by restricting human activity. In contrast, the expansion of illicit crops—often facilitated by illegal armed groups— has been a driver of deforestation (Sánchez-Cuervo et al., 2012; Castro-Nunez et al., 2017a; Ibáñez & Vélez, 2008; Fergusson Talero et al., 2014; Baumann & Kuemmerle, 2016; Morales, 2017; Bautista-Cespedes et al., 2021).

González-González et al. (2021a) find that the effect of armed conflict on deforestation varies with geographic proximity. In the Colombian Amazon, the presence of conflict is positively associated with deforestation in nearby areas; however, this relationship weakens with distance from conflict zones, eventually becoming negative and statistically significant. Similarly, Bautista-Cespedes et al. (2021) show that proximity to both conflict events and mined areas influence deforestation patterns. Their results indicate that (i) areas closer to conflict events tend to experience higher levels of deforestation, whereas (ii) areas closer to mined zones exhibit lower levels of forest loss. Taken together, these findings suggest that the presence of armed groups in the Colombian Amazon may promote deforestation at the local level while potentially reducing it at broader spatial scales, possibly due to processes such as land abandonment.

Several studies indicate that the 2016 peace agreement between the Colombian Government and the FARC-EP led to an increase in deforestation, particularly in the Colombian Amazon. The demobilization of the guerrilla group created opportunities for land grabbers, drug traffickers, other illegal armed groups, ranchers, chieftains, and other actors to access the forest reserve areas within the Amazon biome illegally, including protected areas and Indigenous reservations (Sánchez García & Wong, 2024). In this regard, Murillo-Sandoval et al. (2018) argue that, since the onset of the peace process in 2012 and the signing of the agreement in 2016, there has been a rapid expansion of livestock-driven deforestation beyond the agricultural frontier—even in protected areas. This is consistent with the findings of Clerici et al. (2020), who report a "dramatic and highly significant increase" in deforestation within protected zones.

Beyond conflict intensity itself, the strategies and ideologies of armed actors appear to play a key role in shaping deforestation outcomes. Cantillo and Garza (2022) examine armed conflict in isolation and find that the strategies and ideologies of these actors influence the spatiotemporal patterns of deforestation and conservation. Specifically, municipalities with a stronger presence of leftist guerrillas tend to exhibit lower deforestation rates, whereas a greater presence of right-wing paramilitary groups is associated with higher levels of forest loss. According to Álvarez (2003), guerrilla motivations include, among other factors, strategic considerations. In contrast, the far-right ideology of paramilitary groups is often aligned with the expansion of large estates for livestock production and agribusiness.

2.3. Socio-environmental conflicts and land-use dynamics

Within these structural and conflict-related contexts, several socio-environmental conflicts have emerged as key mechanisms through which deforestation occurs. Among these processes, illegal land grabbing and agricultural expansion have become particularly important drivers of forest loss in the Colombian Amazon (Busch & Ferretti-Gallon, 2023; Armenteras et al., 2013a, 2017; González-González et al., 2021a; Hänggli et al., 2023; Lim et al., 2017; Tebbutt et al., 2021). Land grabbing is often driven by expectations of future land titling, speculative gains in land markets, and the opportunity to consolidate large landholdings at low cost (Sánchez García & Wong, 2024). In many cases, forest clearing serves as a strategy to establish de facto land claims and legitimize property rights.

Land grabbing, along with agricultural expansion for agribusiness and livestock grazing has emerged as one of the most significant drivers of the Amazon's expanding agricultural frontier—surpassing subsistence farming, small-scale agriculture, and even illicit crop cultivation (Murillo-Sandoval et al., 2018; Tebbutt et al., 2021). It is closely linked to the presence of illegal armed groups, road construction, institutional weakness, and the broader absence of effective governance (Tebbutt et al., 2021). According to Sánchez García and Wong (2024), the conversion of forests into pastures is partly driven by the influence of large landowners (latifundistas), who use land accumulation to bolster their social power. This dynamic is further reinforced by the involvement of drug traffickers and other illegal actors who use pasture establishment and livestock farming as mechanisms for money laundering and legitimizing illicit landholding. These activities are commonly used to formalize informal or illegal property claims.

Illicit crop cultivation represents another key socio-environmental process linked to deforestation dynamics in the Colombian Amazon.

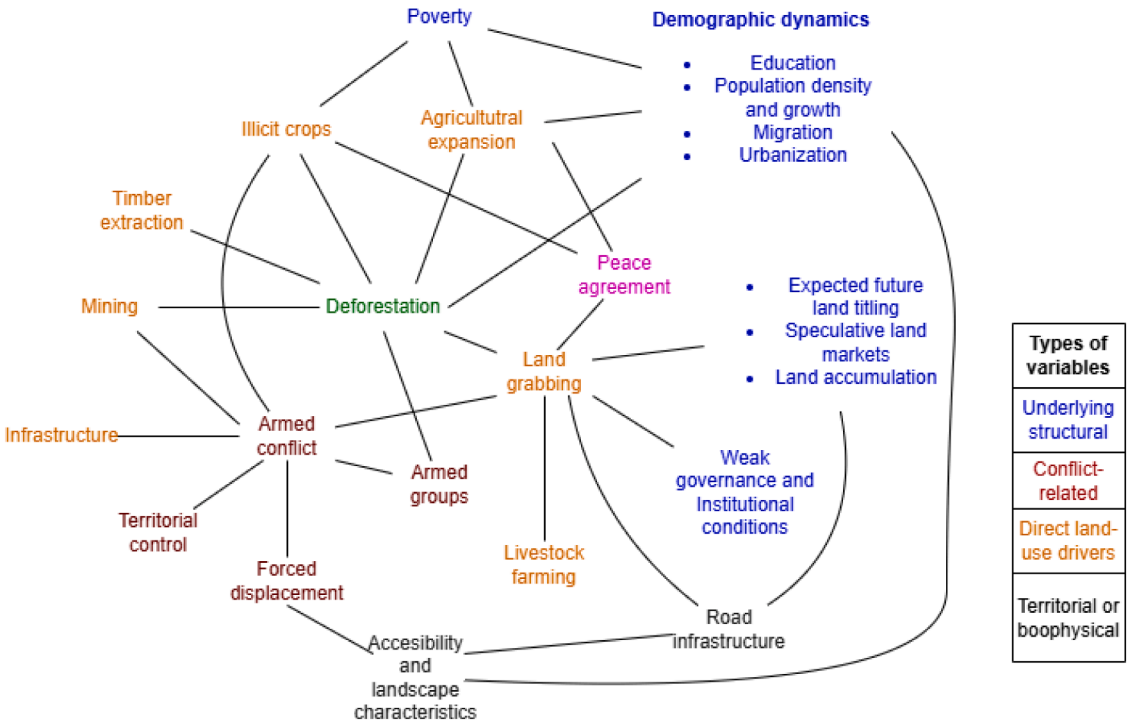

Fig. 1. Conceptual framework of deforestation drivers in the Colombian Amazon.

While several studies identify a significant and direct correlation between the presence of illicit crops and deforestation, others argue that this association is mediated by factors such as armed conflict (e.g., through forced displacement), competing land uses such as pastures, and eradication programs (Dávalos & Dávalos, 2021; Hoffmann et al., 2018; Díaz & Sánchez, 2004; Negret et al., 2019; Dávalos et al., 2009; Rincón-Ruiz & Kallis, 2013). For some authors, the role of illicit crops—particularly coca—is either limited or becomes less pronounced when additional variables, such as demographic indicators, road infrastructure, armed conflict, or illicit crop eradication programs, are included in the analysis (Dávalos et al., 2011, 2016; Hänggli et al., 2023; Negret et al., 2019).

Building on the positive relationship between illicit crop cultivation and deforestation identified by Quiroga-Angel et al. (2022), the authors call for further research into the determinants of illicit crop expansion, particularly its links to factors such as poverty. In this vein, Dávalos and Dávalos (2021) explore the role of illicit crops and population growth as drivers of deforestation within the framework of the impoverishment model, while also identifying infrastructure development as a determinant within the frontier model.

Mining activities also contribute to deforestation in parts of the Amazon, particularly where mining concessions overlap with forest ecosystems and transport infrastructure (González-González et al., 2021b; Hänggli et al., 2023).

2.4. *Territorial accessibility and biophysical conditions*

Territorial accessibility and biophysical conditions further shape the spatial feasibility of these processes. Road infrastructure has played a significant role in deforestation, partly by facilitating migration flows, including those of displaced populations (Dávalos et al., 2016; Hänggli et al., 2023; Hoffmann et al., 2018; Berenguer et al., 2021). Hoffmann et al. (2018) show that road construction is indirectly linked to deforestation through migration and the provision of services, as roads open access to previously conserved areas. Similarly, Mena et al. (2006) note that road construction facilitates market access and promotes colonization and the expansion of the agricultural frontier—a dynamic reflected in the characteristic fishbone pattern (Hänggli et al., 2023). More broadly, expanding road infrastructure also fuels land speculation, strengthens connections with urban centers for the sale of products and the supply of food, inputs, and services, enables the extraction of natural resources such as timber, and raises expectations of land value appreciation.

In addition to infrastructure, biophysical factors such as elevation, proximity to population centers, and distance to existing agricultural frontiers also shape the spatial distribution of deforestation. For example, Bautista-Cespedes et al. (2021) find that deforestation is higher in (i) lowland areas compared to highlands, and (ii) areas closer to population centers. In their stepwise regressions using frontier zone models, elevation, distance to agricultural frontier boundaries, and distance to roads emerge as significant variables.

The literature reviewed underscores the interconnected nature of deforestation drivers in the Colombian Amazon. Structural socioeconomic conditions shape both territorial configurations and conflict dynamics, while armed conflict contributes to the emergence of socio-environmental conflicts such as land grabbing, coca cultivation, and illegal mining. These processes serve as key mechanisms through which land-use transformations and deforestation occur. Territorial and biophysical conditions—including road accessibility and forest fragmentation—further influence the spatial feasibility of these dynamics.

Fig. 1 organizes the relationships identified in the literature into a structured set of mechanisms linking underlying structural conditions, direct drivers of land-use change, conflict-related variables, and territorial conditions. It illustrates how socioeconomic conditions and territorial characteristics shape the emergence and spatial distribution of socio-environmental conflicts, which in turn act as proximate mechanisms driving land-use change and deforestation. By synthesizing these interactions, the framework provides a coherent structure for analyzing how different dimensions interact within the Colombian Amazon. It also serves as the basis for the empirical strategy adopted in this study.

To further synthesize the diverse approaches used in the literature,

Table 1 summarizes key studies on deforestation drivers in the Amazon and the Colombian context. The table highlights the main drivers examined, the geographic scope of each study, and the type of empirical evidence used. This overview illustrates the range of mechanisms proposed to explain deforestation dynamics and underscores the need for integrated approaches that account for interactions between socioeconomic conditions, conflict dynamics, and land-use processes.

Building on this literature, the present study contributes by offering a systematic socioeconomic and demographic characterization of the Colombian Amazon at the municipal level using census data. Beyond a descriptive analysis, it examines how structural socioeconomic conditions are associated with conflict dynamics and socio-environmental processes—such as land grabbing, illicit crop cultivation, and mining—that ultimately shape deforestation outcomes. In doing so, the analysis moves beyond approaches that consider drivers in isolation and instead adopts an integrated perspective that captures the interactions among structural factors, conflict, and land-use dynamics.

3. Data and methods

3.1. Study zone

The Amazon biome in Colombia is in the southern part of the country, bordering Brazil and Peru to the southeast and Ecuador to the south. It encompasses six departments in their entirety—Caquetá, Guaviare, Guainía, Vaupés, Amazonas, and Putumayo—as well as portions of four additional departments: Vichada, Meta, Nariño, and Cauca. In the latter cases, only certain municipalities are partially included within the boundaries of the CAB (Fig. 2).

The CAB comprises 43 municipalities fully within its boundaries, 16 partially included municipalities, and 18 non-municipalized areas. Within this biome, two subregions are distinguished based on their territorial, ecological, demographic, and social characteristics (Guio-Rodriguez & Rojas-Suarez, 2019): the Northwestern Amazon or Andean foothills, and the Southeastern Amazon or lowlands. The Northwestern Amazon is situated in the Andean-Amazonian transition zone and includes the departments of Putumayo, Caquetá, and Guaviare, as well as the Amazonian territories of Meta, Nariño, and Cauca. This subregion covers 37% of the Colombian Amazon's area and is home to 86% of its population. The Southeastern Amazon, or lowlands subregion, includes the departments of Amazonas, Guainía, and Vaupés, along with part of Vichada. It accounts for the remaining 63% of the CAB's area and is inhabited by 14% of its population, predominantly Indigenous communities (Guio-Rodriguez & Rojas-Suarez, 2019).

The Colombian Amazon is home to a rich Indigenous cultural heritage. According to Fundación Alisos (2011), 64 different ethnic groups reside in the area, most of them organized within Indigenous Reserves. Some of these groups are currently considered to be in isolation (Guio-Rodriguez & Rojas-Suarez, 2019; Fundación Alisos, 2011). As of 2024, the DANE Geovisor reports a total of 235 registered Indigenous Reserves in the region.

In addition to the conservation role played by Indigenous Reserves, National Natural Parks (PNN) system constitutes another key conservation mechanism. Approximately 24% of the Amazon biome is designated as National Natural Parks (PNN) (Guio-Rodriguez & Rojas-Suarez, 2019). According to data from the GAIA Amazonas Foundation, these parks cover 10.7 million hectares of the Colombian Amazon.[1] Among them, *Serranía de Chiribiquete* National Park stands out due to its size, spanning 4,268,095 ha.

[1] https://www.gaiaamazonas.org/noticias/2020–07-10_territorios-indigenas-y-areas-protegidas-de-la-amazoniaperdieron-menos-del-1-de-sus-areas-de-bosque-en-treinta-anos/.

3.2. Characterization of the CAB: dimensions, categories, and indicators

The first part of the analysis characterizes the Amazon biome in Colombian across four dimensions:

1. *Social*: Provides a diagnostic overview of population conditions in the biome, using demographic, social, economic, and institutional indicators.
2. *Forest cover*: Describes the condition and degree of degradation of terrestrial ecosystems, with a focus on the forest cover.
3. *Freshwater*: Presents indicators reflecting both the natural availability of water and human demand for freshwater-associated ecosystem services.
4. *Socio-environmental Conflicts*: Captures conflicts arising from human activities that affect terrestrial and freshwater ecosystems in the Amazon biome.

Specific analytical categories are defined for each of these dimensions, each comprising multiple indicators. These categories are illustrated in Fig. 3.

The social dimension includes demographic, socioeconomic, and institutional indicators, presented and analyzed at the municipal and departmental levels for 2018 or 2019, depending on data availability. These are grouped into six categories, as shown in Fig. 4.

Data for the social dimension was obtained from the following sources: (i) the 2018 National Census (population, migration, and poverty indicators), (ii) the National Department of Statistics – DANE (economic activity indicators), (iii) the Gran Encuesta Integrada de Hogares (GIEH) (unemployment), (iv) UPRA- Ministry of Agriculture (agricultural information), (v) Ministry of Transport (roads and airports), (vi) Min-TIC (internet access), and (vii) the National Department of Planning – DNP (institutional performance).

Because some municipalities are only partially included in the CAB and sub-municipal data are unavailable, certain values were calculated using weighted approaches. Indicators in the Population and Agricultural Activity categories were weighted by municipal areas, while those for Migration, Municipal Value Added, and Digital Connectivity were weighted by population. Poverty and institutional performance indicators reflect values for the complete municipality. Other spatially based indicators—such as road density and airport locations—were derived directly from maps produced by IDEAM- SINCHI.

The forest dimension comprises three categories: natural cover, deforested area, and forest fragmentation (Fig. 5). Indicators for these categories are derived from maps produced by IDEAM-SINCHI. Data are available for the years 2002 to 2020; to ensure consistency with census information, data from 2019 were used.

In this case, municipal estimates are constructed by aggregating the pixel-level data from the maps for each indicator across municipalities wholly or partially included in the CAB.

Forest cover is measured in two ways: (i) the absolute area in hectares (forest area) and (ii) the proportion of the departmental or municipal territory covered by forests (percentage of municipal area under forest). Deforestation and fragmentation are measured using three metrics: (1) absolute area in hectares (deforested or fragmented areas); (2) the share of these areas relative to the total forest area within the municipality or department (percentage of deforestation or of fragmented forest); and (3) the share relative to the total surface area of the municipality or department (percentage of municipal area deforested or fragmented). Because forest cover accounts for a large share of the biome's total area, the latter two measures tend to yield similar aggregate values. However, at the municipal level, more pronounced differences may emerge.

For the freshwater dimension, the National Water Status Report (ENA in Spanish) provides a comprehensive set of indicators on the country's water conditions (Instituto de Hidrología, Meteorología y Estudios Ambientales (IDEAM), 2023). For this study, selected ENA

Table 1
Summary of key studies on deforestation drivers in the Colombian Amazon.

Study	Region	Main drivers	Type of evidence	Key insight
Armenteras et al. (2006)	Colombia	Structural, land-use	Correlational	Early identification of deforestation drivers
Murad & Pearse (2018)	Colombia	Poverty	Econometric	Poverty as underlying driver
Hoffmann et al. (2018)	Colombia	Coca, roads, poverty	Econometric	Indirect effects via migration
Dávalos et al. (2016)	Colombia	Coca, migration	Correlational	Coca linked to forest loss
Bautista-Cespedes et al. (2021)	Colombia	Conflict, infrastructure	Econometric	Conflict and spatial drivers
Clerici et al. (2020)	Amazon	Conflict (post-peace)	Empirical	Increase in deforestation after peace
Murillo-Sandoval et al. (2020)	Colombia	Conflict	Empirical	Conflict may reduce deforestation
Prem et al. (2020)	Colombia	Conflict	Causal	Armed conflict increases deforestation
Fergusson Talero et al. (2014)	Colombia	Conflict	Causal	Economic mechanisms of conflict
Cantillo & Garza (2022)	Colombia	Armed actors	Causal	Ideology affects deforestation
Armenteras et al. (2013a, 2017)	Colombia	Agriculture	Remote sensing	Frontier expansion drives deforestation
González-González et al. (2021a)	Colombia	Conflict	Econometric	Local vs regional effects
González-González et al. (2021b)	Colombia	Mining	Econometric	Mining affects forest loss
Dávalos and Dávalos (2021)	Colombia	Coca, poverty	Conceptual + empirical	Impoverishment model
Hänggli et al. (2023)	Amazon	Roads, population, mining	Econometric	Integrated drivers' model

indicators were extracted and organized into the categories presented in Fig. 6. Within the Amazon biome, IDEAM defines 11 hydrological zones and 57 subzones. In the Amazon Basin, these subzones may span multiple municipalities, and a single municipality may encompass one or more subzones. To derive municipal-level estimates, indicator values reported by IDEAM (2023) for each subzone were weighted proportionally to the share of each municipality's area within them. Some of the indicators were adjusted to ensure comparability, allowing all to be expressed on a categorical scale from 0 to 4, corresponding to very low, low, medium, high, and very high levels. In all cases, lower values indicate more favorable conditions.

The indicators are calculated and analyzed at both municipal and departmental levels for 2022, based on the most recent report published by IDEAM. While they do not directly assess the status of aquatic ecosystems, they provide an approximation of the freshwater provision services.

The first category, natural freshwater supply, examines subzone dynamics in terms of their capacity to generate freshwater over time and the potential threats to that capacity. The second category, anthropogenic demand for freshwater, analyzes how human activities affect freshwater provision in the subzones, including population needs, habitat transformation, and changes in water quality. Finally, the third category—the interaction between demand and supply— assesses the pressures on freshwater systems when both variables are considered jointly.

The socio-environmental conflicts dimension encompasses four categories commonly identified in literature as drivers of forest loss in the CAB: land grabbing, illicit-crop cultivation, mining, and armed conflict and violence (Fig. 7).

Data for this dimension were drawn from multiple sources: (i) SINCHI-IDEAM (deforestation and fires), (ii) the UNODC Illicit-use Monitoring System (illicit crop cultivation), (iii) the National Mining Agency and RAISG (legal and illegal mining), (iv) INDEPAZ (presence of armed groups), (v) the Victims Unit (expulsion and reception of displaced populations), and (vi) the Ministry of Defense (acts of war, homicides, and homicide rates).

For municipalities that are only partially included, the indicators for land grabbing, illicit crop cultivation, and mining are based on spatial data corresponding to the portion of the municipality within the CAB. In contrast, indicators for the presence of armed groups, acts of war, and homicide rates are assigned using values for the municipality. Indicators related to the expulsion and reception of displaced populations are weighted by population. As with the other indicators, the most recent data available, closest to 2018, is used.

While some variables correspond to directly observed phenomena (e.g., coca cultivation or conflict events), others—such as land grabbing—are approximated using observable indicators, reflecting the inherent difficulty of measuring complex socio-environmental processes. Given the strong association between land grabbing, pasture conversion, and fires, these variables are used as proxies.

3.3. Interactions

We conduct a Pearson correlation analysis across all pairs of included variables, including both indicators, and categories. The analysis focuses on interactions that are statistically significant ($p\text{-value} < 0.01$) and those that (i) are correlated with deforestation and (ii) involve variables that are themselves significantly associated with deforestation. Some of the relationships suggest preliminary signs of causality, which inform the development of an econometric model to identify the drivers of deforestation in the Colombian Amazon.

Given the scope of the dataset compiled in this study—approximately 305 variables across different dimensions, categories, and indicators for the years 2015 to 2020—some relationships may not be fully captured by the main methodologies employed. While the primary objective is to examine the link between deforestation and socio-environmental conflicts, we complement this approach with econometric and machine learning (ML) tools to uncover patterns and interactions among the variables in the CAB region, helping to identify the drivers and channels through which they operate.

3.4. Econometric model

The empirical strategy developed in this study builds on the construction of a comprehensive and internally consistent dataset for the Amazon biome in Colombia, integrating indicators across four dimensions: socioeconomic and demographic conditions, socio-environmental conflicts, freshwater supply and demand, and terrestrial ecosystems. This data framework enables a coherent representation of complex regional dynamics in a context marked by limited and fragmented information.

The literature reviewed (Table 1) shows a strong association between deforestation and various socio-environmental conflicts, including armed conflict, land grabbing, coca cultivation, and mining. These

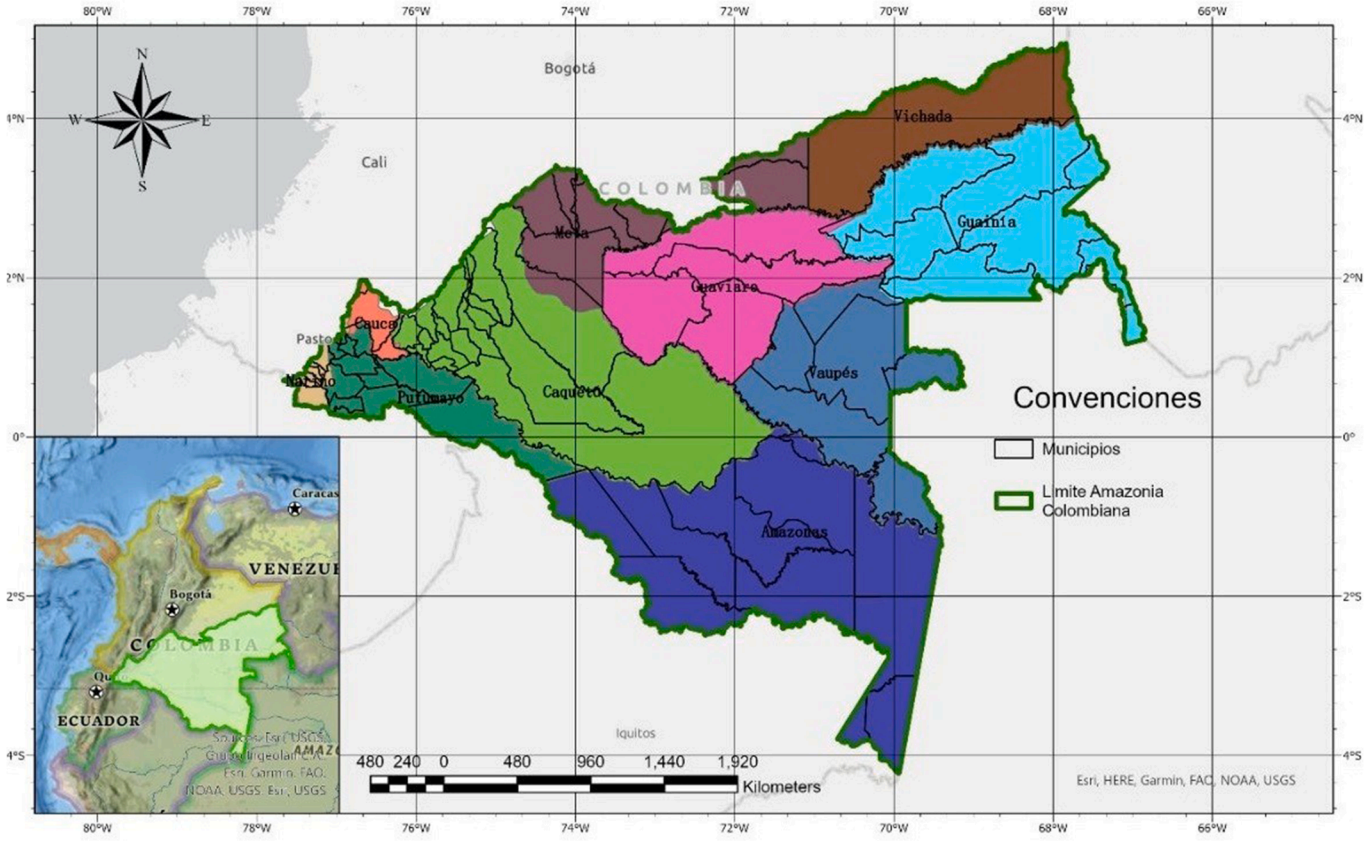

Fig. 2. The Colombian Amazon Biome, and its departments, and municipalities.

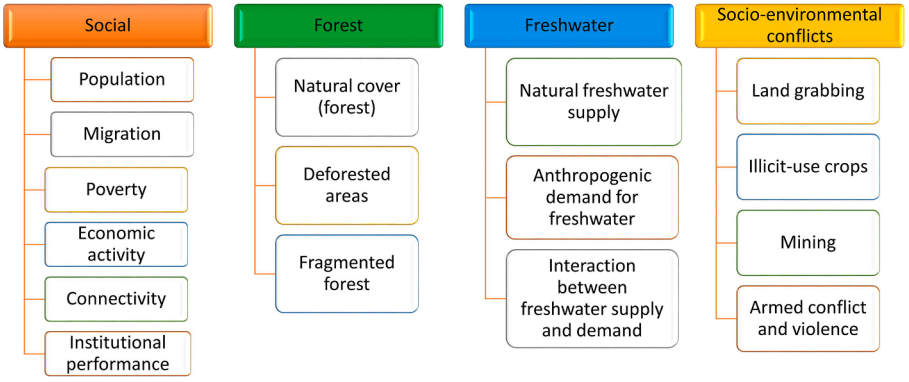

Fig. 3. Dimensions and categories for assessing the status of the Amazon Biome.

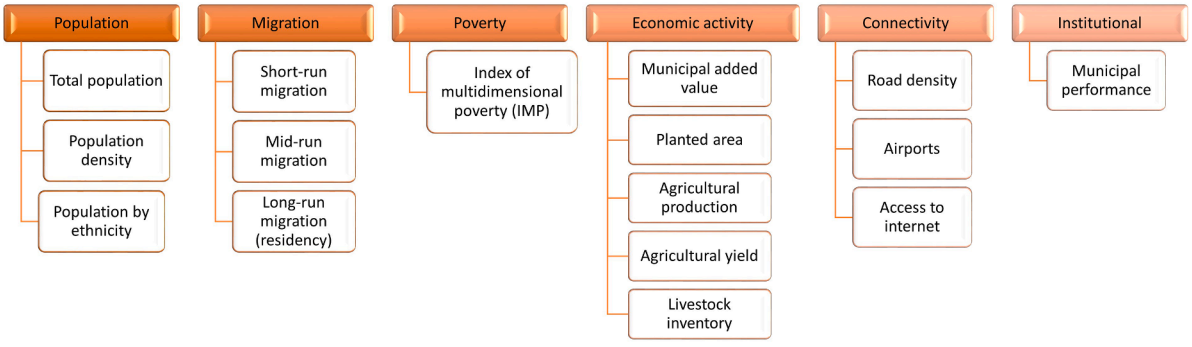

Fig. 4. Categories and indicators for the social dimension.

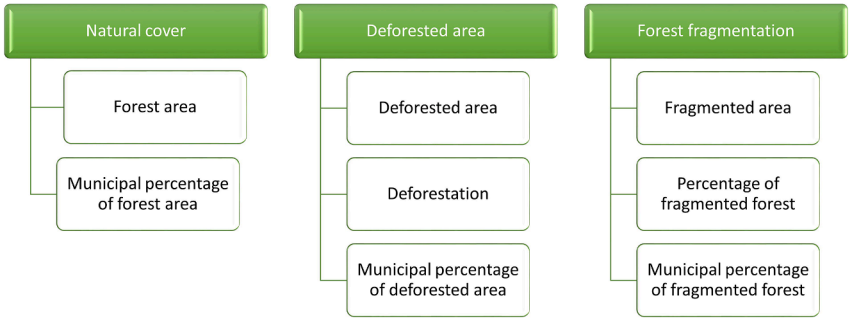

Fig. 5. Categories and indicators for the forest dimension.

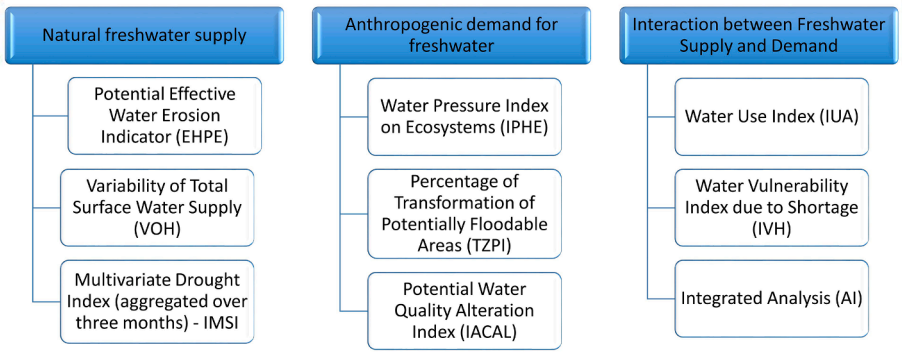

Fig. 6. Categories and indicators for the status of the freshwater dimension.

Fig. 7. Categories and indicators for the socio-environmental conflicts dimension.

conflicts are, in turn, closely linked to socioeconomic conditions and to the decisions that drive deforestation. Within this framework, econometric models and machine learning techniques are used jointly to analyze interactions among indicators and identify robust correlational patterns associated with deforestation. Rather than operating as separate analytical tools, these approaches are integrated into a unified strategy that connects data construction with empirical analysis, enabling a more systematic examination of how multiple, interrelated factors contribute to forest loss.

3.4.1. *Structural model*

The empirical analysis is based on a cross-sectional dataset at the municipal level. The dependent variable measures deforestation observed in 2019 while explanatory variables correspond to the most recent information available prior to, or contemporaneous with, 2019 depending on the reporting cycle of each data source, mostly 2018. The temporal reference for each variable is summarized in Table 2. The following equation formalizes the structural relationship between different forms of socio-environmental conflict and deforestation:

$$Deforestation_i = \alpha + \beta_1 LandGrabbing_i + \beta_2 CocaCultivation_i + \beta_3 Mining_i + \gamma ArmedConflict_i + \delta X_i + \mu_i$$

Where X_i includes socioeconomic and territorial controls—such as road infrastructure, poverty indicators, and the share of Indigenous population—at each administrative territory. The dependent variable is measured in hectares rather than as a share of municipal area to capture the absolute magnitude of forest loss. We recognize that, as stated by Baland and Libois (2026), estimating the relations in shares produces variances that are more independent of the area, in the Colombian Amazon Basin, however, municipality sizes vary substantially, causing proportional measures to be driven largely by differences in

Table 2
Descriptive statistics of the main variables used in the model.

Variable	Unit	Period	Mean	Std. Dev.	Min	Max
Deforested area	Hectares	2019	1,474	3,030	10.64	17,439
Post-FARC armed groups	Intensity	2017–2019	0.41	0.83	0	3
Land grabbing	Hectares	2016–18	5,502	11,489	0	55,275
Coca cultivation	Hectares	2018	634.3	1,221.3	0	7,616.2
Illegal mining	Count / intensity	2018	2.60	8.36	0	57
Migration	Share	2018	0.41	0.22	0.08	0.80
Type 1 Roads	Km	2018	6.68	16.31	0	91.96
Type 3 Roads	Km	2018	1.19	2.39	0	12.95
Type 5–7 Roads	Km	2018	314.7	452.3	0	2,667
Forest cover	Municipal share	2018	0.66	0.29	0.02	1.03
Fragmented Forest	Municipal share	2018	0.02	0.02	0.0002	0.09
Indigenous population	Municipal share	2018	0.39	0.36	0.001	0.99
Poverty	Index	2018	59.59	22.67	16.30	98.50
Zone	Dummy (0/1)	—	0.65	0.48	0	1

Notes: Statistics are computed at the municipality level. The dependent variable corresponds to deforestation observed in 2019. Explanatory variables are measured using the most recent information available prior to the outcome depending on the reporting cycle of each data source, mostly 2018. Post-FARC armed group presence is constructed as an intensity measure ranging from 0 to 3, where 0 indicates municipalities with no armed group presence during the 2017–2019 period, and 3 indicates municipalities where armed groups were present in all three years analyzed. Migration measures the share of residents who declared having changed municipality during the five years preceding the 2018 Population Census (2013–2018). Land grabbing is measured as cumulative pasture expansion during 2016–2018. Shares range between 0 and 1. Missing observations are minimal (1 out of 78 for most variables). Road variables represent total length by type. The variable zone distinguishes between the northwestern (1) and eastern (0) regions of the Colombian Amazon Basin.

administrative area rather than by the absolute magnitude of forest loss. Therefore, normalizing deforestation by municipal area would instead measure deforestation intensity, thereby addressing a different empirical question. Moreover, given the substantial variation in municipality size across the Colombian Amazon Basin, such normalization would make the outcome largely reflect differences in administrative area rather than variation in forest loss. Differences in the initial availability of forest are instead accounted for through the inclusion of baseline forest cover and forest fragmentation in the empirical specifications.

This specification, together with the evidence reviewed in the literature, points to a clear endogenous relationship between several socioeconomic and conflict-related variables and deforestation, making causal inference challenging. Any model intended to identify the drivers of deforestation in the CAB must therefore account for this endogeneity, as these variables both influence land-use decisions and are shaped by underlying socioeconomic conditions.

3.4.2. Two-stage model

One variable that plays a key role in both deforestation dynamics and socio-environmental conflicts is armed conflict. However, the pathways through which it affects deforestation remain unclear. We propose treating variables associated with armed conflict as factors that help characterize the emergence of other socio-environmental conflicts, rather than as direct determinants of deforestation. At the same time, we do not rule out a direct relationship between armed conflict and forest loss, including through mechanisms such as aerial bombardment, forced displacement and land abandonment, war economies not mediated through formal land grabbing, or shifts in state enforcement capacity. For instance, Bautista-Cespedes et al. (2021) show that proximity to both conflict events and mined areas influence deforestation patterns.

However, given the complexity of these channels and their potential overlap with broader territorial dynamics, we treat such direct effects as conceptually plausible but empirically difficult to disentangle. Instead, our empirical strategy focuses on isolating the component of socio-environmental conflicts that are systematically associated with conflict dynamics, capturing the extent to which armed conflict may operate through these intermediate processes. This approach allows us to distinguish between proximate land-use mechanisms and the broader conditions that may give rise to them, without imposing strong assumptions about the absence of direct effects. A visual representation of the model is provided in Fig. 8.

To account for the elements outlined above, we estimate a system of equations using two-stage least squares (2SLS) to capture the joint determination of deforestation and key land-use processes. This approach allows us to represent the conceptual channels linking conflict, land-use dynamics, and forest loss.

In this setup, the first stage estimates each socio-environmental conflict as a function of armed conflict and a set of channel-specific socioeconomic and territorial variables. Based on key findings in the literature and the analysis of available data, this stage is represented by the following equations:

$$Landgrabbing_i = f_1(armedconflict_i, roads_i, Migration_i, fragmentedforest_i, indigenouspopulation_i, poverty_i, region_i)$$

$$Cocacultivation_i = f_2(armedconflict_i, roads_i, indigenouspopulation_i, poverty_i, region_i)$$

$$Mining_i = f_3(armedconflict_i, roads_i, naturalcover_i, indigenouspopulation_i, poverty_i, region_i)$$

In the second stage, the predicted values of conflict variables from the first stage are used to explain deforestation, along with a set of additional controls. Because each socio-environmental conflict may be driven by distinct underlying factors, the second stage is specified accordingly. As a result, we estimate three separate equations:

$$Deforestation_i = g_1(\widehat{landgrabbing}_i, migration_i, fragmentedforest_i, indigenouspopulation_i, poverty_i, region_i)$$

$$Deforestation_i = g_2(\widehat{cocacultivation}_i, indigenouspopulation_i, poverty_i, region_i)$$

$$Deforestation_i = g_3(\widehat{mining}_i, naturalcover_i, indigenouspopulation_i, poverty_i, region_i)$$

We model armed conflict primarily as a channel-shifting variable that operates through socio-environmental conflicts. This strategy is best understood as a partial endogeneity-correction mechanism rather than a fully identified structural causal model.

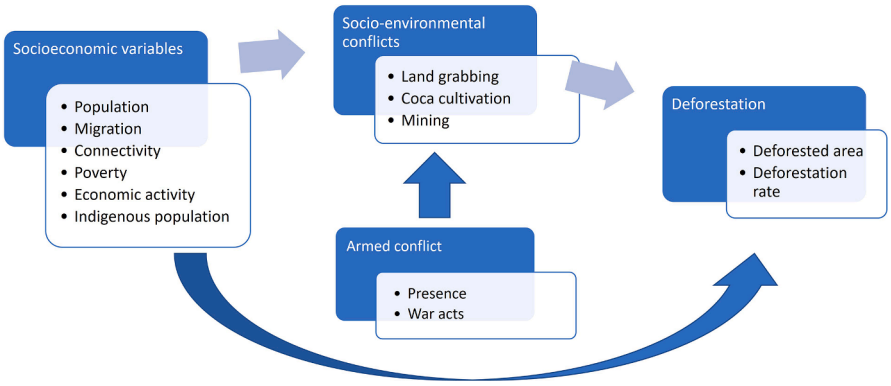

Fig. 8. Graphical model of the relationships between deforestation, socioeconomic variables, and socio-environmental conflict variables.

3.4.3. Variable inclusion criteria

The selection of variables across specifications follows a conceptual rationale, complemented by an exploration of interactions among indicators—both within and across dimensions—to assess their relevance and suitability for inclusion (See Section 2 of the Supplementary Material). Socioeconomic controls (e.g., poverty index, Indigenous population share) are included in the second-stage equations to account for structural determinants of land-use change.

In all first-stage regressions, a common set of structural controls (armed conflict, Indigenous population, poverty, and zone) is included to account for baseline socioeconomic and regional heterogeneity. Additional regressors differ across equations because they represent mechanism-specific determinants identified through the conceptual and empirical selection procedure described below.

Road infrastructure and baseline environmental conditions (forest cover or forest fragmentation) are incorporated selectively across first- and second-stage equations according to the theoretical mechanism under consideration. Variables expected to influence the emergence of socio-environmental conflicts are included in the first stage, whereas structural determinants of deforestation are retained in the second stage. While freshwater supply and demand indicators were initially considered as part of the broader set of environmental covariates, they are not retained in the final specifications. Their inclusion was assessed alongside other variables using both conceptual criteria and regularization-based selection procedures, as described below.

This distinction reflects our attempt to separate structural covariates from channel-specific predictors while maintaining consistency with the mediation framework depicted in Fig. 8.

Baseline forest cover or forest fragmentation is included in the empirical specifications whenever theoretically relevant to account for differences in the amount and condition of forest available for conversion. These variables are preferred to municipality areas because they capture the ecological conditions underlying deforestation processes rather than differences in administrative size. Table 2 presents summary statistics for the main variables used in the analysis. Deforestation varies substantially across municipalities, with a mean of 1,474 ha and a wide dispersion, indicating high heterogeneity in forest loss across the Colombian Amazon. Land-use pressures also differ considerably: pasture expansion and coca cultivation exhibit skewed distributions, with many municipalities showing low levels and a few exhibiting very high values. Infrastructure variables—particularly secondary and tertiary roads—are unevenly distributed, reflecting differences in accessibility. Socioeconomic indicators such as poverty and migration also vary widely, while environmental variables point to high average forest cover alongside relatively low but uneven levels of fragmentation. Overall, the data reveal pronounced spatial heterogeneity across municipalities in both the drivers and conditions associated with deforestation.

3.4.4. Machine learning procedure

Given the high dimensionality of the covariate space, the relatively small sample size, and the potential for measurement error, the baseline specification may be prone to overfitting and instability. To address this, we implement LASSO-based variable selection procedures to identify a parsimonious set of predictors. LASSO is used as a regularization tool to reduce dimensionality, rather than as a standalone predictive method.

First, within the 2SLS framework, we estimate the model using a subset of variables selected via LASSO. This approach is particularly useful in settings with potentially correlated predictors and a relatively small cross-sectional sample, as it enables automatic variable selection while reducing overfitting. The regularization parameter (λ) is selected via k-fold cross-validation to minimize out-of-sample prediction error.

We then employ gradient boosting methods (XGBoost) to assess the relative importance of predictors and test the robustness of the relationships identified in the econometric models. XGBoost accommodates nonlinear relationships and higher-order interactions among covariates that may not be captured by linear specifications. Hyperparameters are tuned via cross-validation, and model performance is evaluated using out-of-sample error metrics.

These machine learning exercises provide complementary evidence on the predictive relevance and relative importance of socioeconomic and socio-environmental conflict drivers of deforestation. The convergence of findings across econometric and nonlinear boosting models strengthens the robustness of the empirical patterns. Further details are provided in the Supplementary Material.

4. Results

4.1. Indicators

This section presents an overview of the dimensions, categories, and indicators developed in this study. Detailed results for each indicator are provided in the Supplementary Material.

4.1.1. Social dimension

Population

According to Colombia's 2018 Census, the Amazon Biome region has a relatively small population compared to the rest of the country, with just over one million people—about 2.3% of the national total. Slightly more than half of the CAB's population resides in municipal capitals (51%), compared to 77% nationwide. This distribution has remained largely unchanged since 2012, according to figures reported by Trujillo-Quintero (2014).

Given that the CAB covers nearly half of Colombia's land area, its low population density is not surprising (Fig. 9). While the national average is approximately 39 people per km^2, the Amazon Biome averages just 4.5.

The CAB's population pyramid differs from the national profile, with

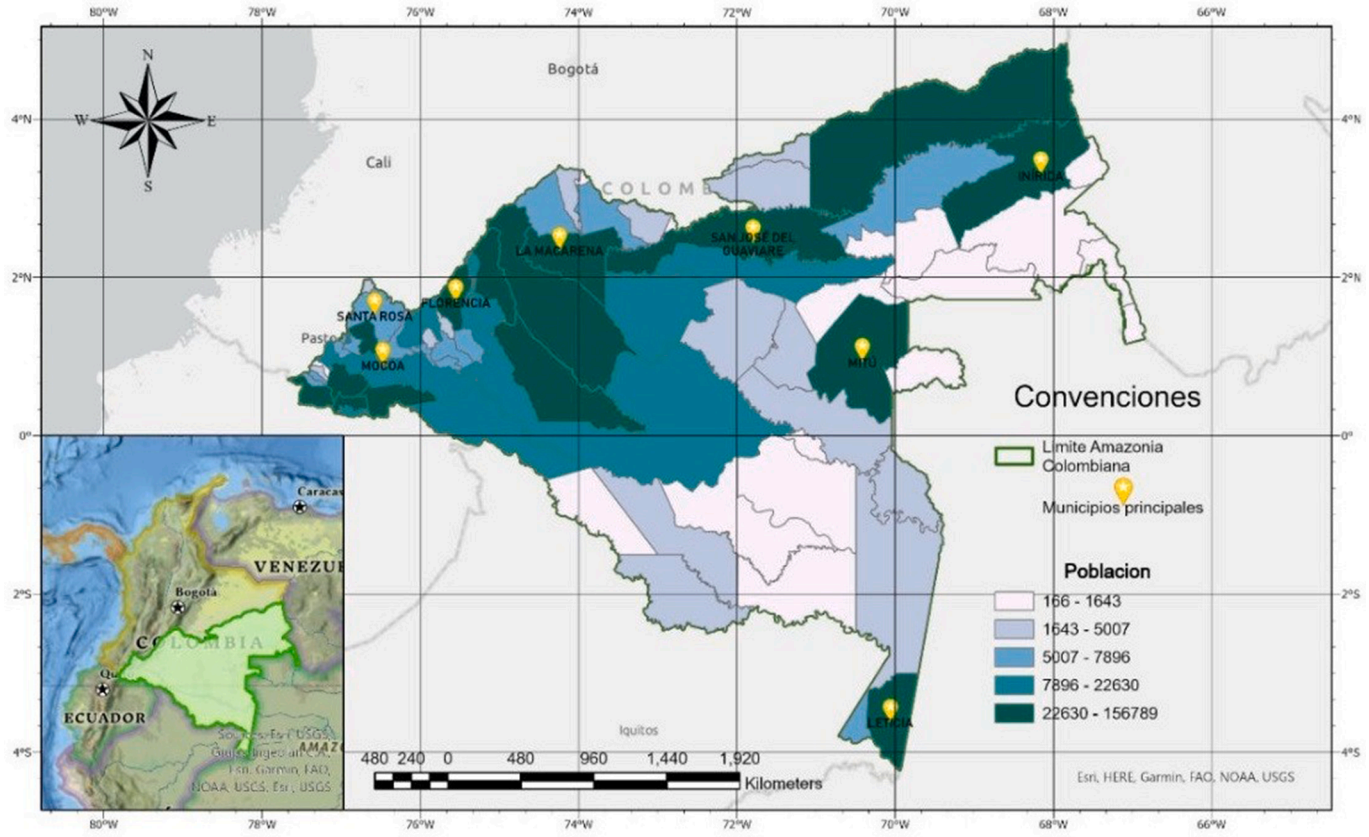

Fig. 9. Total population by municipality in the Amazon Biome, 2018. .
Source: Authors based on CNVP – 2018, DANE (2018)

a significantly higher share of individuals under 25. Nearly 50% of the Amazon Biome's population falls within this age group, indicating a predominantly young demographic.

The CAB is home to a significant Indigenous population, with a wide range of ethnic groups across its territory. According to the 2018 Census, 224,627 inhabitants identify as Indigenous—22% of the region's population. This contrasts sharply with the national figure, where Indigenous peoples account for just 4.2% of the population. Notably, the CAB includes 12% of the country's total Indigenous population, underscoring the strong presence of ethnic groups in the Amazon region. The eastern departments—Vaupés, Guainía, Amazonas—as well as the Amazonian portion of Vichada, have the highest shares of Indigenous residents relative to their total populations. Together, these areas are home to 54% of the CAB's Indigenous population.

Migration

According to the 2018 Census, 36,419 people migrated to the CAB that year, accounting for 3.6% of the region's population. In the five years preceding the census, 98,659 people relocated to municipalities within the CAB—equivalent to 9.7% of the total population. In other words, roughly one in ten CAB residents had lived elsewhere in the previous five years. The residence indicator reveals that, on average, about one-third of the CAB's population was not born in the municipality where they lived at the time of the 2018 Census—a proportion slightly higher than the national average (26%).

Poverty

In terms of multidimensional poverty, most municipalities and departments in the Amazon region face precarious conditions. While the national average stands at 19%, the mean rises to 60% in the CAB.

Notably, in the departments of Vaupés, Guainía, and Amazonas, nearly 90% of the population lives in conditions of multidimensional poverty. By contrast, departments in the Northwestern Amazon Arc (Putumayo, Caquetá, and Guaviare) report poverty rates below the CAB average (Fig. 10).

Connectivity

According to INVIAS, the Amazon biome contains 26,530 km of roads, accounting for 6.9% of the country's total road network (Table S-1 in Supplementary Material). The CAB's road network is marked by limited development of secondary roads and the predominance of tertiary roads (Fundación Alisos, 2011). Most roads are concentrated in the northwestern arc, with Caquetá alone accounting for half of the biome's total road infrastructure (13,347 km).

The CAB contains just 514 km of primary roads, accounting for only 2.5% of the national total. Across all CAB departments, the road network is dominated by Type 6 and 7—unpaved and unimproved paths and trails. These account for the largest share of roads in the Amazon biome, totaling over 19,000 km, or 73% of the biome's network and 8.5% of this road type nationwide.

The limited road network across most of the CAB means that transport relies primarily on river routes—where navigable waterways exist—or on-air travel.

Considering all road types, the national average road density is 0.336 km per km^2. By contrast, the Amazon biome averages just 0.055 km/km^2— approximately one-sixth of the national average.

Another key dimension of regional connectivity is digital access. Nationally, Colombia averages 17 fixed internet service subscriptions per 100 inhabitants, corresponding to a penetration rate of 0.17. In the Amazon biome, however, this figure falls to 0.08—roughly half the national average.

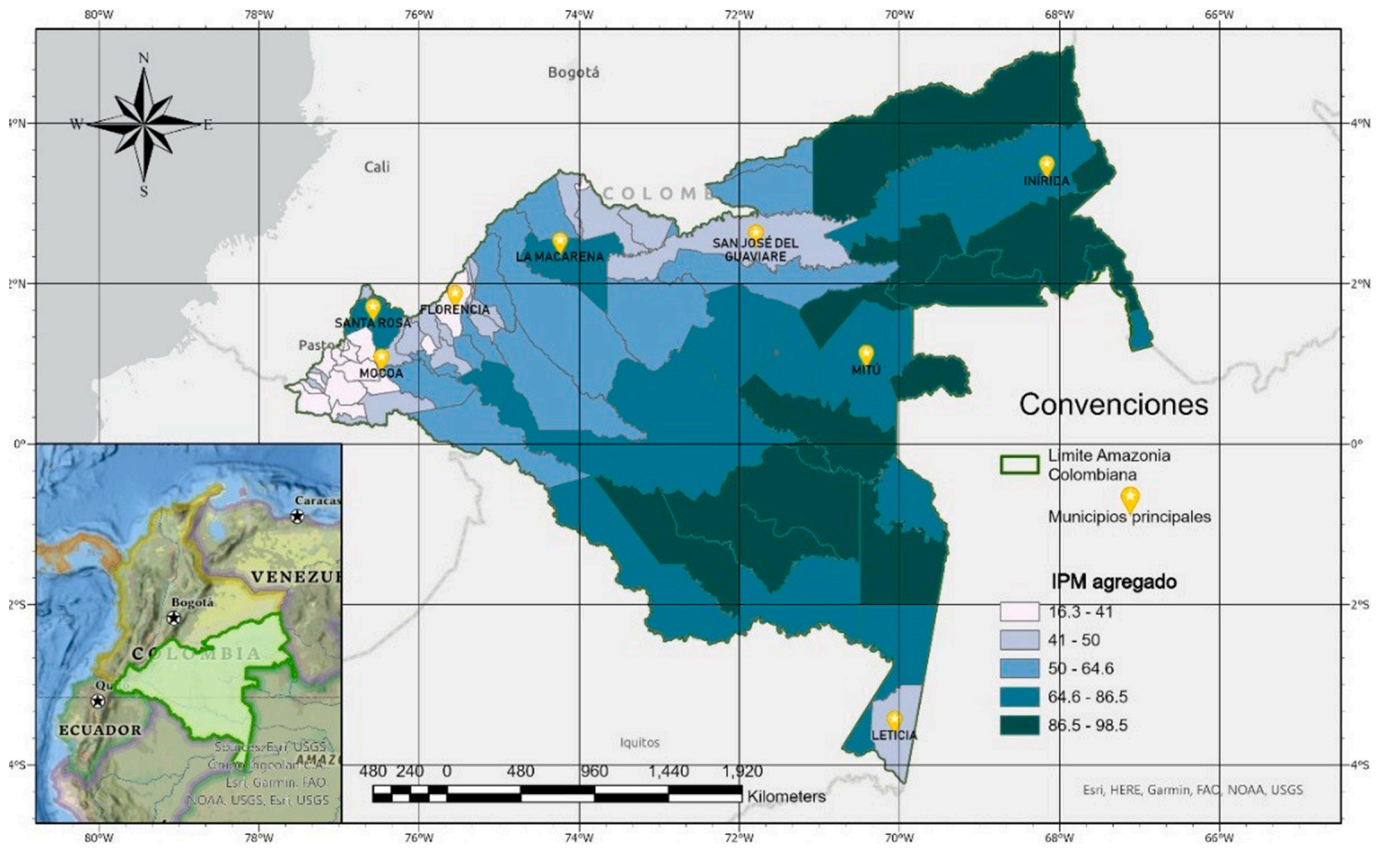

Fig. 10. Multidimensional poverty index by municipalities of the Amazon Biome, 2018. Source: Authors based on DANE (2023).

4.1.2. Forest-cover dimension

Natural Cover

Between 2002 and 2022, natural forest area in the Amazon biome declined from 41.46 million hectares to 39.17 million hectares—a loss of 2,228,144 ha over two decades, averaging 114,000 ha per year. This represents a 5.5% reduction in forest cover. Notably, forest loss accelerated in the second decade, with a 3.1% decrease (from 40.4 to 39.2 million hectares), compared to a 2.5% reduction in the first decade (from 41.5 to 40.4 million hectares).

As of 2018, an estimated 82% of the CAB remained covered by natural forests. In 23 territories, forest conservation rates—measured as a percentage of total land area— exceeded 90%. However, in ten municipalities, most of them in Caquetá, forest cover accounted for less than one-quarter of the land.

Deforested area

Deforestation is the primary threat to the Amazon biome, driving land-use change, biodiversity loss, and the degradation of vital ecosystem services. Forest loss is often followed by the adoption of productive practices poorly suited to the biome's ecological conditions. According to data from IDEAM and SINCHI, an average of 109,000 ha

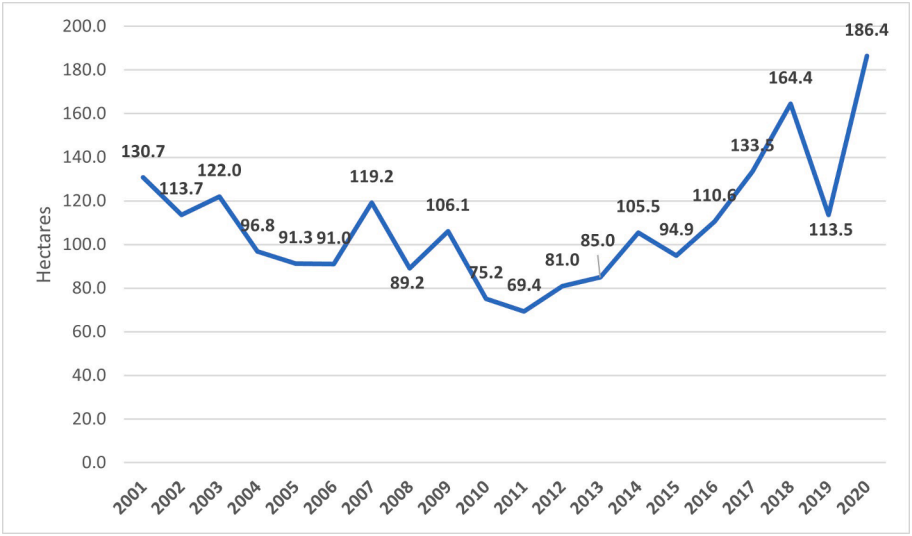

Fig. 11. Annually deforested areas in the Amazon Biome from 2001 to 2020. Source: Authors based on IDEAM-SINCHI.

were deforested annually between 2001 and 2020—amounting to a total of 2,179,465 ha by 2020, or equivalent to 4.51% of the Amazon biome.

Between 2001 and 2020, deforestation followed a cyclical pattern shaped by external factors. As shown in Fig. 11, deforested areas declined during the first decade of the century, but reversed course in the second, with a marked acceleration and peaks in 2018 and 2020. Notably, the COVID-19 pandemic appears to have further intensified deforestation in the CAB (Fiestas-Flores, Higa & Montoya-Zumaeta, 2025).

These findings align with Botero (2018), who notes that deforestation in the Colombian Amazon has followed a steady—though cyclical—pattern. He also highlights an uptick during the peace agreement period. In areas with a longstanding guerrilla presence, even where sensitive zones underwent significant transformations, deforestation tended to be more limited and localized. However, since the peace agreement, state efforts to assert territorial control in these areas have been, at best, slow.

While deforested areas provide an initial indication of forest cover loss in the Amazon biome (Fig. 12), the most widely used metric to assess this process is the deforestation rate, defined as the deforested area relative to the forest area during the reference period. Between 2001 and 2020, the Amazon biome lost 4.4% of its forest area. Like the deforested area, this indicator also shows some variations over time.

Fig. 13 shows a decline in the annual deforestation rate from 2002 to 2012, followed by an increase through 2016. In 2018, the rate declines slightly. Fig. 14 presents municipal-level deforestation rates for 2018–2019 in the CAB.

Forest fragmentation

Forest fragmentation is a form of intervention that typically precedes deforestation and can therefore serve as a predictive indicator. In the Amazon biome, an average of approximately 550,000 ha has been affected by fragmentation during the 21st century, although this figure has fluctuated over time.

In 2018, fragmented forest accounted for 1.26% of the Amazon biome's natural forest. The most heavily fragmented areas are in Amazonian Cauca, Putumayo, Amazonian Meta, and Amazonian Nariño, indicating that the Amazonian piedmont is disproportionately affected by this phenomenon.

4.1.3. Socio-environmental conflicts dimension

This dimension includes four categories: land grabbing, illicit crop cultivation, mining, armed conflict and violence.

Land grabbing

Land grabbing primarily targets inalienable state-owned territories—such as forest reserves, national parks, and Indigenous lands—which are illegally appropriated. As a socio-environmental process, land grabbing is not directly observable and therefore cannot be captured by a single indicator. Instead, it is understood as a sequence of actions that begins with forest clearing (felling and burning), often accompanied by road construction. This is followed by the introduction of productive activities such as cattle ranching or crop cultivation, including coca. Finally, by demonstrating occupation and "improvements," claims to ownership are asserted, enabling transactions despite the land having been acquired illegally (Instituto Igarapé and InSight Crime, 2021).

Land-grabbing takes multiple forms and is therefore approached empirically through observable indicators that capture different stages of the process. In this study, it is analyzed primarily through indicators of pasture expansion and, to a lesser extent, burning. Pasture expansion—the conversion of other land uses into pastures—is a key phenomenon in land grabbing in the Amazon biome. It is also closely associated with cattle ranching, which has developed under an extensive production model. This model is characterized by low investment, free grazing, limited soil and pasture renewal, and low productivity per animal and per unit of land (Ruiz-Santacruz (2022), as cited in Fundación Alisos (2011)).

Although these indicators should not be interpreted as direct measures of land grabbing, they are highly correlated with land appropriation dynamics in the Amazon. Between 2002 and 2020, an average of 194,000 ha were converted to pasture each year in the CAB. Fig. 15 illustrates the intensity of pasture expansion across CAB municipalities during the 2018–2020 period.

According to studies by RAISG, FCDS, and SINCHI, burning was recorded across 2,072,165 ha. The most severely affected departments were Caquetá, accounting for 39% of the total burned area, and Amazonian Meta with 33%, followed by Guaviare at 17%.

Illicit crop cultivation

Illicit crop cultivation is primarily associated with forest loss in the Amazon's forest reserves, national parks, and Indigenous territories. However, the literature suggests that these crops are not the main driver of deforestation (Erasso & Vélez, 2020).

Illicit coca cultivation in the CAB began in the departments of Caquetá, Putumayo, and Guaviare in the 1980s and has since become a source of income for illegal armed groups (Trujillo-Quintero, 2014). The

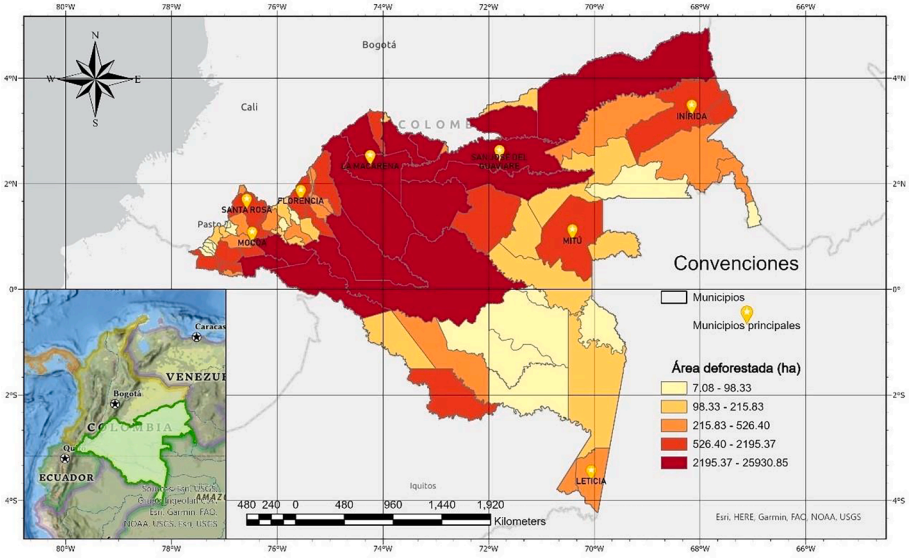

Fig. 12. Deforested area at the municipal level in the CAB during 2018. Source: Authors based on SIAC-SINCHI.

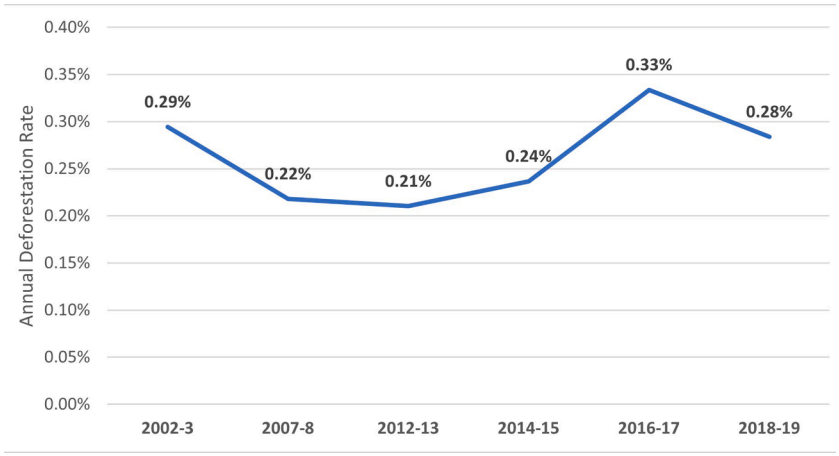

Fig. 13. Annual deforestation rates in the Amazon Biome from 2002 to 3 to 2018–9. Source: Authors based on SIAC-SINCHI.

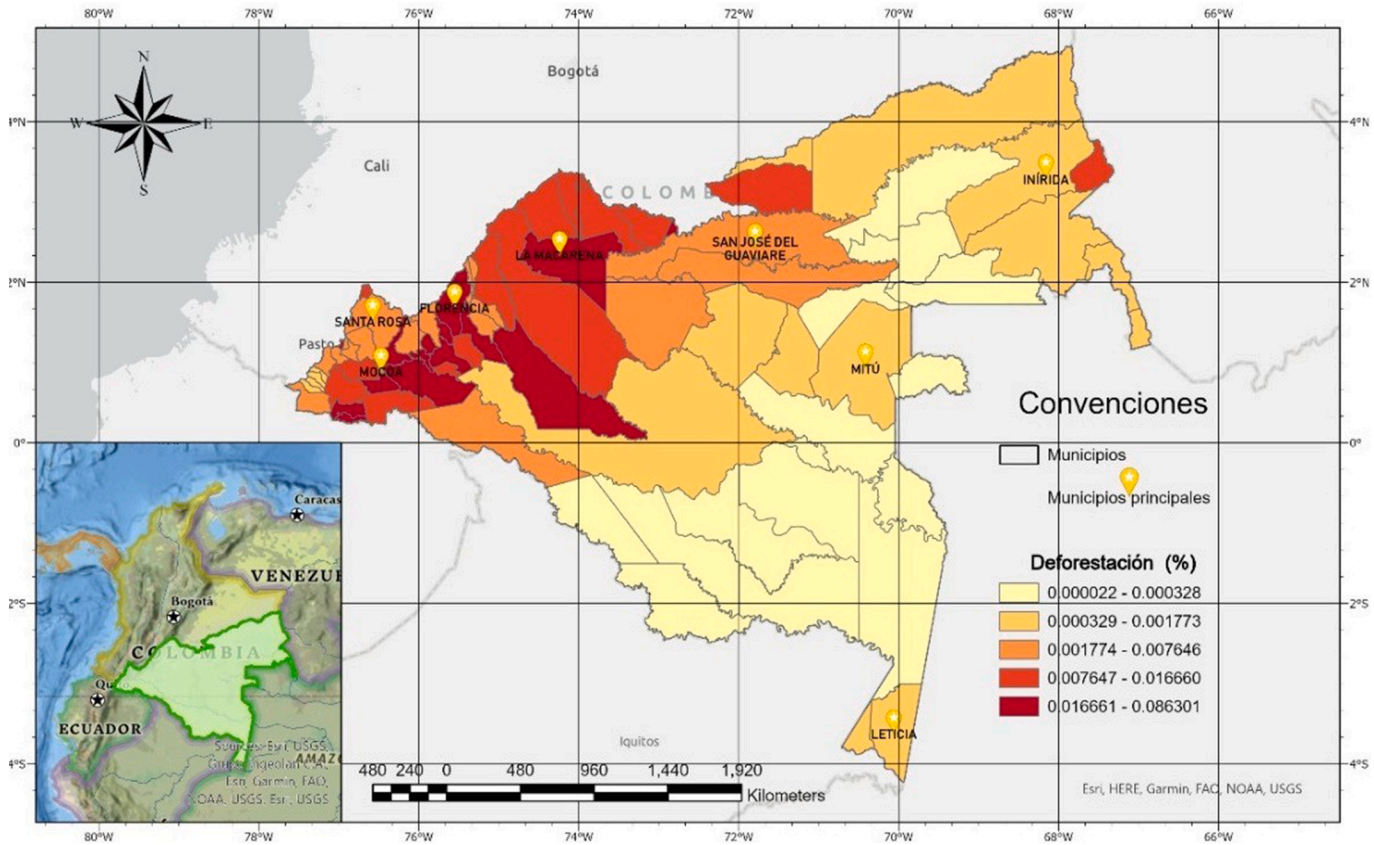

Fig. 14. Municipal-level deforestation rates in the CAB for 2018–2019. Source: Authors based on SIAC-SINCHI.

area under coca cultivation in the Amazon biome has fluctuated over time. From nearly 100,000 ha in 2001, it declined sharply during the first decade of the century to approximately 16,600 ha by 2010, likely reflecting a shift of coca production to other regions.

From 2010 onward, however, cultivated areas rebounded steadily, peaking between 2012 and 2017 before declining again by 2020. By 2022, coca cultivation had surged to 65,000 ha (Fig. 16).

The expansion of coca cultivation in national parks is particularly notable. Guio-Rodriguez and Rojas-Suarez (2019) report that between 2016 and 2017, *Sierra de la Macarena* PNN was the most severely affected, with 2832 ha under coca. During the same period, *Alto Fragua Indi Wasi* PNN experienced a 96% increase in coca cultivation; however, by 2024, coca crops in this park had almost entirely disappeared. Coca cultivation declined in *La Paya* PNN and the *Nukak Reserve* between 2016 and 2017, due to forced manual eradication efforts. Fig. 17 shows the extension of coca cultivation at municipal level in the CAB during 2018.

Mining

Mining is an economic activity that often affects the ecosystems in which it takes place. In the Amazon biome, a wide range of minerals have been identified for extraction and commercialization, making mining a common—rather than exceptional—practice in the region. Numerous concessions and permits have been granted, primarily for the extraction of construction materials. At the same time, the presence of valuable minerals such as coltan or gold has increased pressure on the biome, often through illegal extraction activities. These practices further

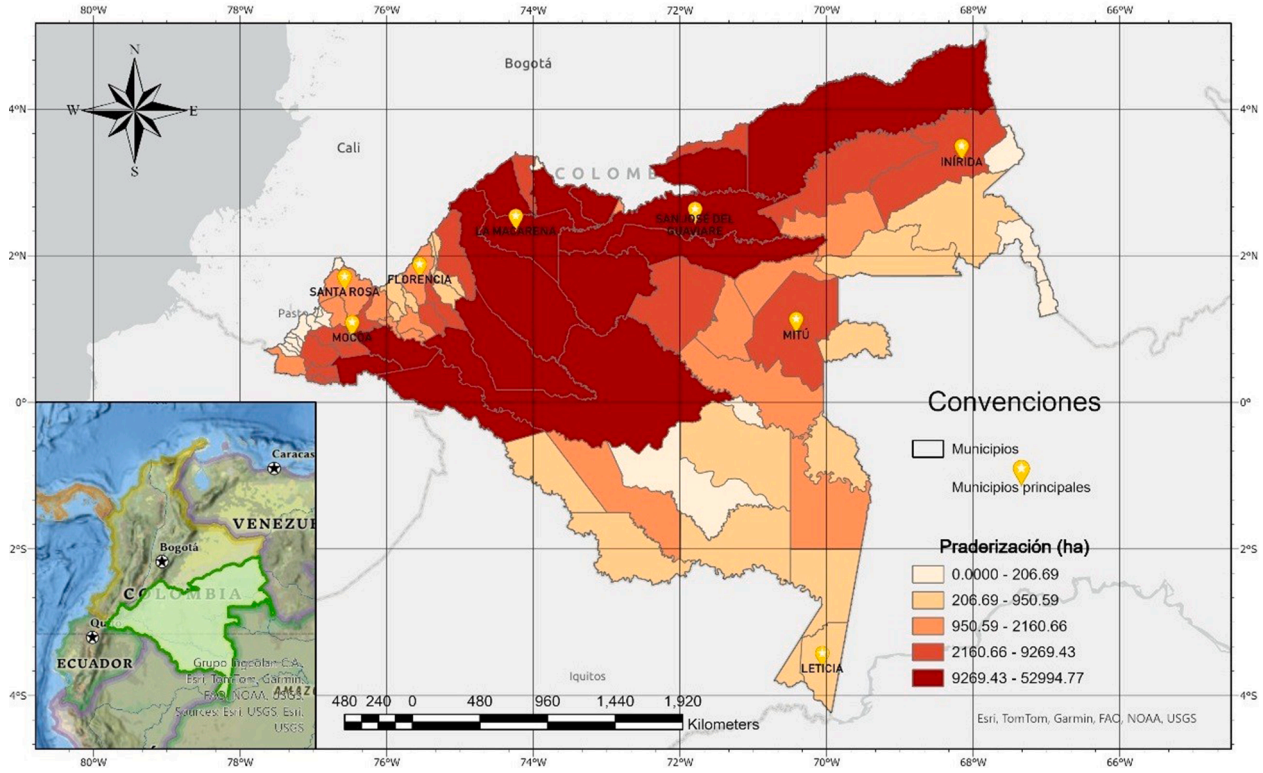

Fig. 15. Land grabbing in the 2018–2020 period in CAB municipalities. Source: Authors based on SIAC-SINCHI.

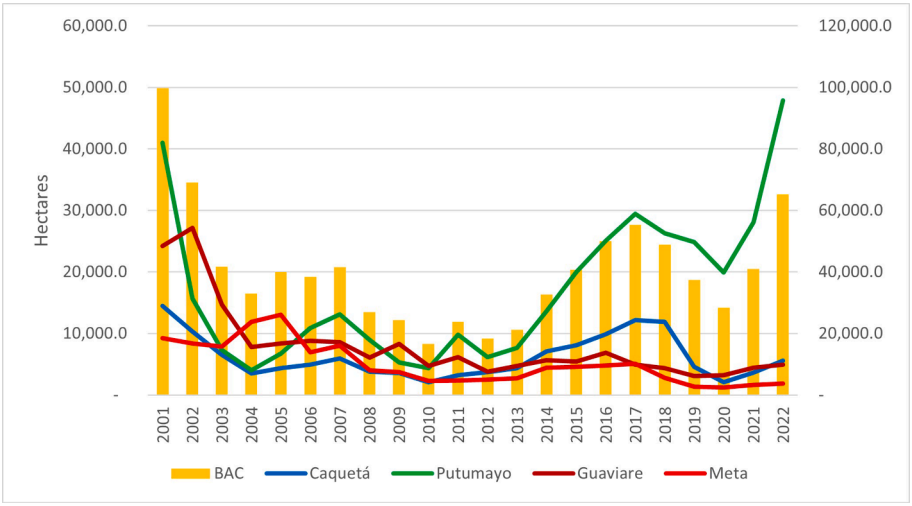

Fig. 16. Evolution of area planted with coca in the CAB and in the main departments. . (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)
Source: Authors based on SIMCI, UNODC Note: The right axis represents the value for the entire CAB, shown as the overall average in the yellow bar

degrade environmental conditions, particularly in contexts of weak monitoring and enforcement. Beyond gold, the mineral wealth of the Colombian Amazon includes cadmium, copper, cobalt, tin, iron, and molybdenum, among others (United Nations Office on Drugs and Crime, UNODC 2020). Fundación Alisos (2011) also identifies, in order of importance, the presence of tantalum, niobium, vanadium, zirconium, platinum, construction materials, titanium, and rutile.

Illegal mining activities affect the health of local populations—primarily Indigenous communities—due to the release of mercury and other contaminants. Water and soil quality deteriorate, largely because of erosion; natural capital is degraded; governance within Indigenous territories is weakened, along with the capacity to develop strategies for economic sustainability (Guio-Rodriguez & Rojas-Suarez, 2019). In addition, Fundación Alisos (2011) highlights other mining-related impacts, including the removal of vegetation cover; forest degradation; the alteration of watercourses, riverbeds, and banks; water acidification; and increased turbidity. These effects compromise the integrity of protected areas, Indigenous territories, and the lands of peoples in voluntary isolation. Specifically, the authors estimate that producing one kilogram of gold requires 1.3 kg of mercury and the removal of one cubic meter of soil for every gram of gold produced.

The sharp increase in gold prices in recent years has made it a major source of income for illegal groups, which rely on unsustainable extraction techniques and do not regulate the disposal of toxic

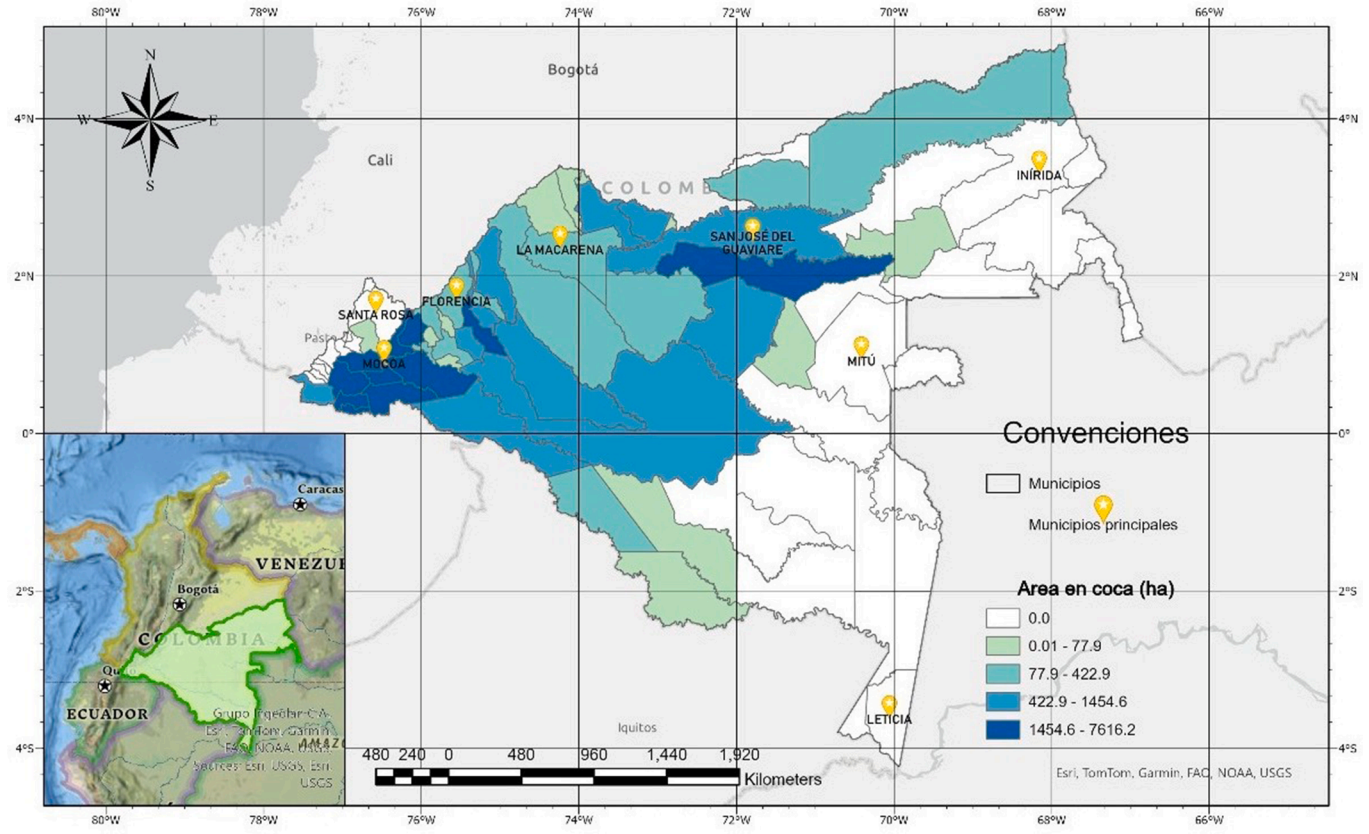

Fig. 17. Area planted with coca crops at the municipal level in the CAB during 2018. Source: Authors based on SIMCI-UNODC

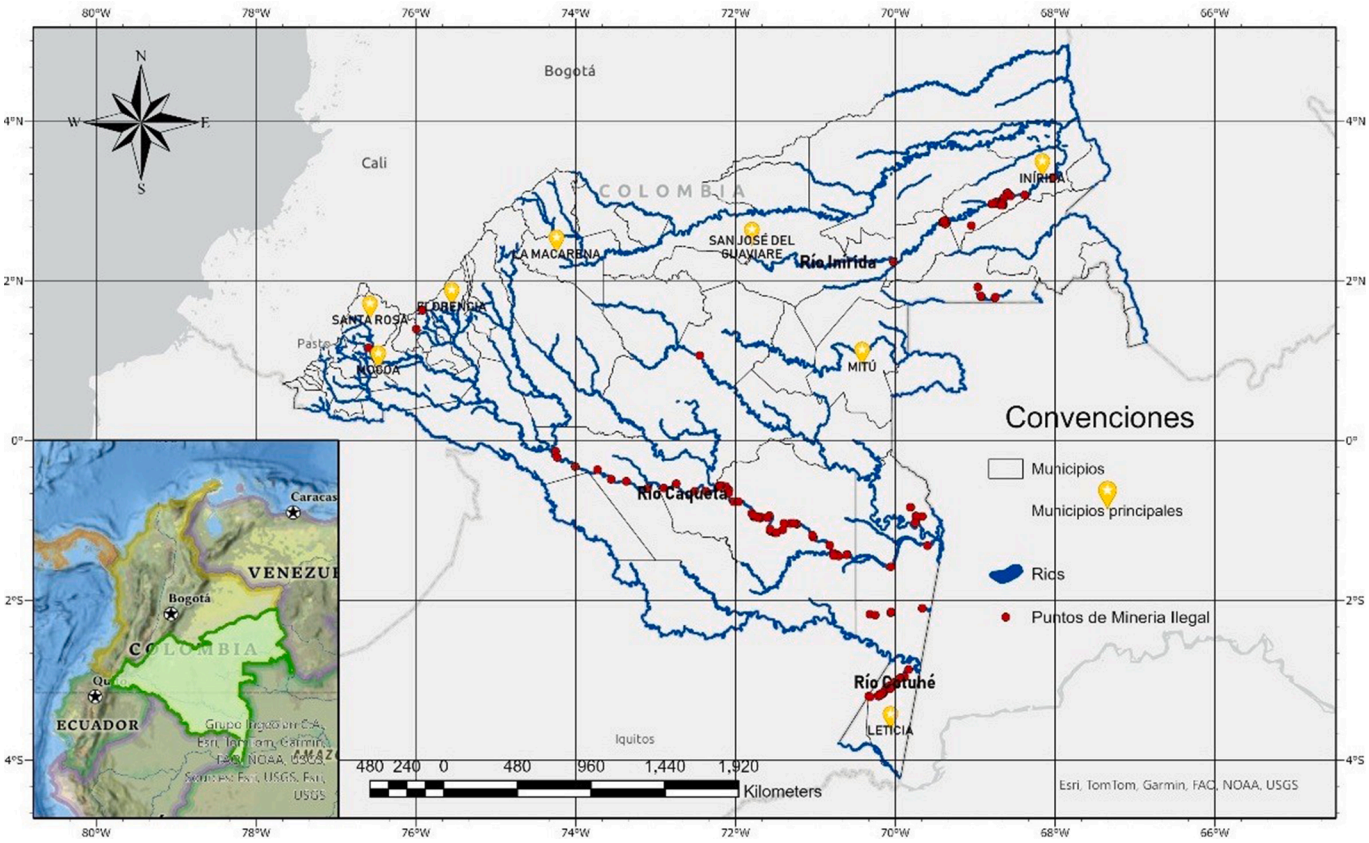

Fig. 18. Reported points of illegal mining in the CAB – 2020. Source: Authors based on RAISG.

substances such as mercury and cyanide used in the process. According to Suárez et al. (2021), gold exports have surpassed those of cocaine, generating approximately two billion dollars annually.

Informal mining refers to operations that are not legal but are undergoing a formalization process; they do not take place in prohibited areas and do not use unauthorized machinery. In contrast, illegal mining occurs in restricted zones and/or fails to comply with administrative, technical, social, and environmental legal requirements (MINAM (2013) cited by FCDS (2024)).

Alluvial mining typically occurs illegally along riverbanks (Suárez et al., 2021). According to Botero (2018), illegal mining is present in major and mid-sized rivers across the Colombian Amazon, including the Guaviare, Inírida, Guainía, Isana, Vaupés, Apaporis, Tararira, Caquetá, Putumayo, Orteguaza, Cotohué, Puré, Yarí, and Caguán. The SINCHI Institute identifies five key areas where illegal mining is concentrated: 1) Caquetá – Putumayo, 2) Caquetá River, 3) Cotohué River; 4) Vaupés; and 5) Guainía. Fig. 18 presents reported points of illegal mining in the CAB, corresponding to 200 sites identified by RAISG for 2020.

Each extraction point typically operates between one and eight machines. A single dredge can produce around 40 g of gold per day—approximately 14 kg per year—generating between US $150,000 and $200,000 in income (Instituto Igarapé and InSight Crime, 2021). However, the illicit nature of these activities makes it difficult to obtain comprehensive data, and these figures may underestimate the true scale of mining in the region.

Armed conflict and violence

Armed conflict and associated violence have affected the entire country, albeit unevenly. In the Amazon region, two opposing dynamics converge. On one hand, the region's remoteness makes it a strategic location for illegal armed groups, which face less pressure from security forces and can more easily transport weapons and illicit substances. Activities such as coca cultivation and cocaine production, unregulated mining, and other profitable ventures further increase the region's appeal for these groups. On the other hand, this same isolation makes some areas difficult to access, reducing their geopolitical significance. Together, these factors generate sustained pressure on the biome from armed groups, manifesting in acts of war, civilian displacement, and both direct and indirect harm to human life.

Regarding the presence of armed groups, the literature indicates that conflict dynamics in the region changed significantly following the 2016 peace agreement with the FARC-EP. Areas previously controlled by this guerrilla group were left without effective governance, and the state failed to fill the resulting power vacuum. In their absence, the ELN guerrilla, criminal gangs (BACRIM), remnants of paramilitary groups, and especially FARC-EP dissidents moved into these vacated territories (Instituto Igarapé and InSight Crime, 2021). Collectively, these actors are referred to as post-FARC armed groups (PFAG). According to Indepaz, PFAG operated in 28 municipalities across the Amazon biome in 2019 (Fig. 19).

Another way to understand the patterns of violence in the region is through reported military actions. In 2015, 66 such actions were recorded across the Amazon biome. This number dropped sharply to just four in 2016, likely reflecting the peace agreement. However, in 2017, it rose again to levels like those of 2015 (64), before declining steadily to a low of five in 2019 (5), with a slight uptick in 2020 (7).

One of the main consequences of violence and armed conflict is the forced displacement. Fig. 20 illustrates the impact of peace agreement on this phenomenon. In 2013 and 2014, approximately 30,000 people were displaced from their municipalities in the CAB due to violence. By 2016, this number had fallen to fewer than 11,000. In 2017 and 2018, displacement stabilized at around 7,000 to 8000 people per year, with a slight increase in the latter year.

Another indicator of violence affecting the population is the number of homicides. Between 2003 and 2008 the Amazon biome experienced high homicide rates, with over 1000 deaths per year. This was followed by a decline to around 800 homicides annually between 2009 and 2013, and a further drop to approximately 500 deaths per year beginning in 2016.

According to SINCHI (2023), Putumayo is the CAB department with the highest number of homicides of social leaders, human rights defenders, and peace agreement signatories. The same source reports 94 such killings between 2016 and 2022. Additionally, 14 massacres were recorded in Putumayo between 2020 and 2022.

In 2018, Colombia's national homicide rate was 2.93 per 10,000 inhabitants, while the rate in the Amazon biome was slightly higher, at 3.25.

4.2. Interaction between indicators

Correlation analysis helps reveal how social indicators and socio-environmental conflicts are associated with indicators of ecosystem conservation. This provides a foundation for developing models to explore causal relationships or determinants of deforestation.

First, municipalities with larger total populations and those with

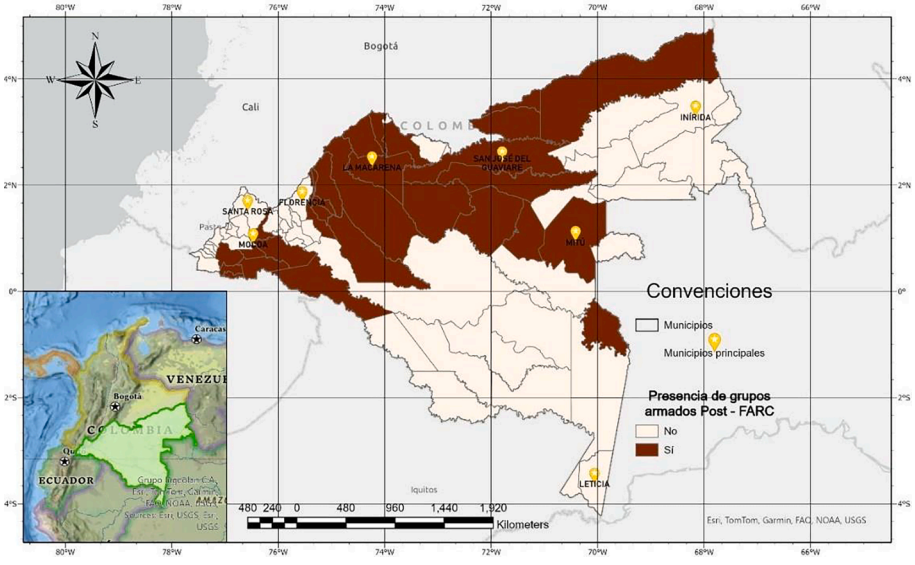

Fig. 19. Municipalities in the Amazon Biome with the presence of Post-FARC armed groups in 2019. Source: Authors based on USAID database.

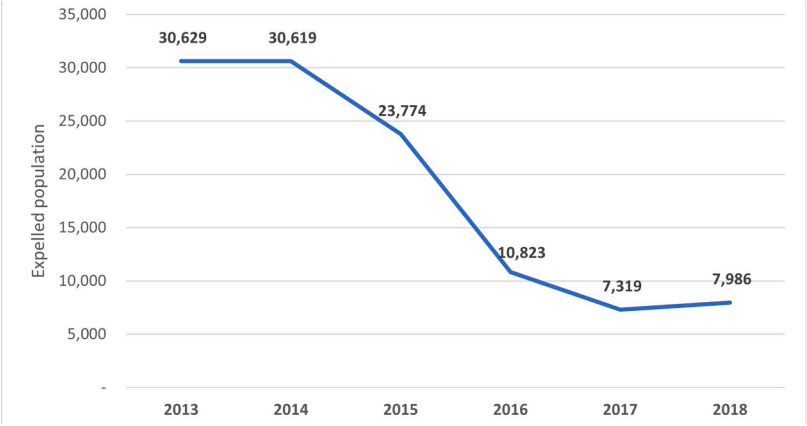

Fig. 20. Population expelled from their municipalities due to displacement in the amazon biome from 2013 to 2018. .
Source: Registro Único de Víctimas based on Datos CEDE, CEDE

higher shares of Indigenous residents show contrasting patterns across socioeconomic indicators. More populous municipalities tend to have higher levels of migrants, lower poverty levels, greater economic activity, larger areas under permanent crops and higher production, larger cattle inventories, more extensive road infrastructure, better internet access, and stronger institutional performance. By contrast, municipalities with higher Indigenous populations display the opposite pattern: fewer migrants, higher poverty, reduced economic activity, smaller areas under permanent crops and lower production, smaller cattle inventories, fewer roads, poorer internet access, and weaker institutional performance (Fig. 21).

Municipalities with a greater presence of post-FARC armed groups are characterized by higher levels of displacement, more frequent military actions, and higher homicide rates. These municipalities also tend to be associated with areas experiencing greater land grabbing (pasture expansion and burning) as well as a stronger presence of illicit coca cultivation (Fig. 22).

Conversely, the presence of permanent crops, livestock inventories, and road infrastructure is positively associated with land grabbing, the presence of armed groups and military actions, population displacement, and homicides. While livestock activity is more strongly associated with land grabbing, road infrastructure is more closely linked to armed groups, displacement, and homicides.

Specifically, roads classified as types V, VI, and VII play a more prominent role in these dynamics. Municipalities with higher migration rates and stronger economic performance tend to be associated with socio-environmental conflicts such as land grabbing, armed conflict, and homicides. Conversely, a higher share of Indigenous population is associated with lower levels of all these conflicts. At the same time, both legal and illegal mining activities are more prevalent in municipalities with larger Indigenous populations and higher levels of poverty (Fig. 23).

Deforestation indicators across municipalities in the Amazon biome are positively associated with various socio-environmental conflicts, except for mining. Municipalities experiencing higher levels of deforestation tend to have a greater presence of armed groups, more frequent military actions, higher levels of displacement, and more homicides. Higher deforestation is also linked to increased land grabbing and the presence of coca crops. Similarly, forest fragmentation is positively associated with pasture expansion, coca cultivation, and violence related to armed groups and homicides, though not with mining activity. By contrast, the presence of natural forests is associated with municipalities that have fewer coca crops and homicides, but higher levels of mining activity. Selected additional correlations are reported in the Supplementary Material.

Building on IDEAM (2018), Instituto Igarapé and InSight Crime

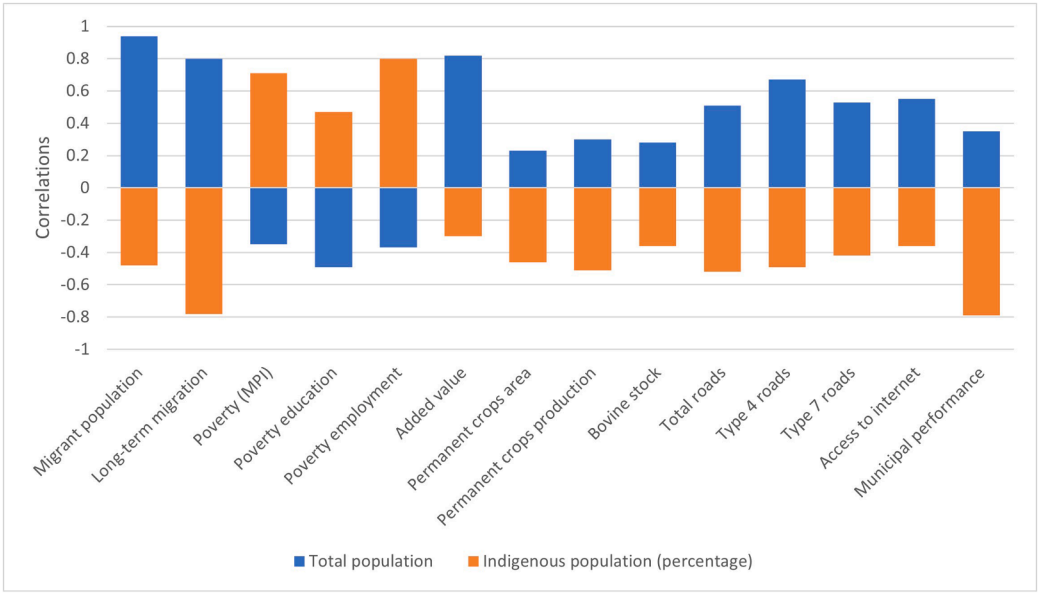

Fig. 21. Main correlations of total population indicators and percentage of indigenous population with other social indicators.

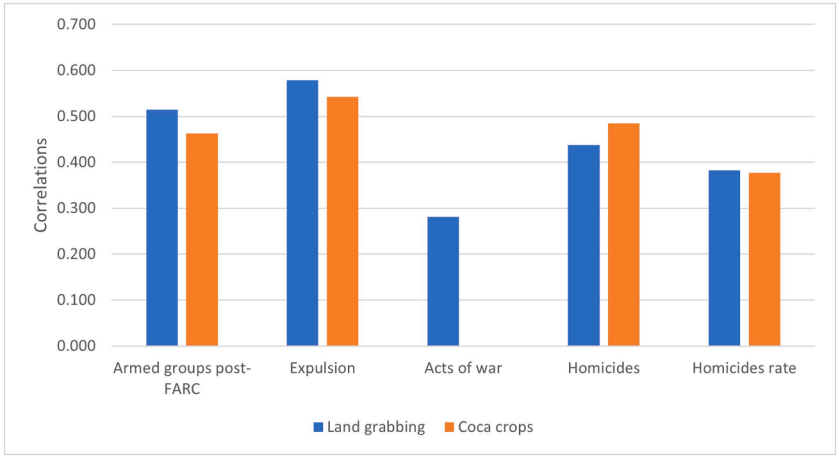

Fig. 22. Main correlations between land grabbing and the presence of illicit crops indicators and conflict indicators associated with violence.

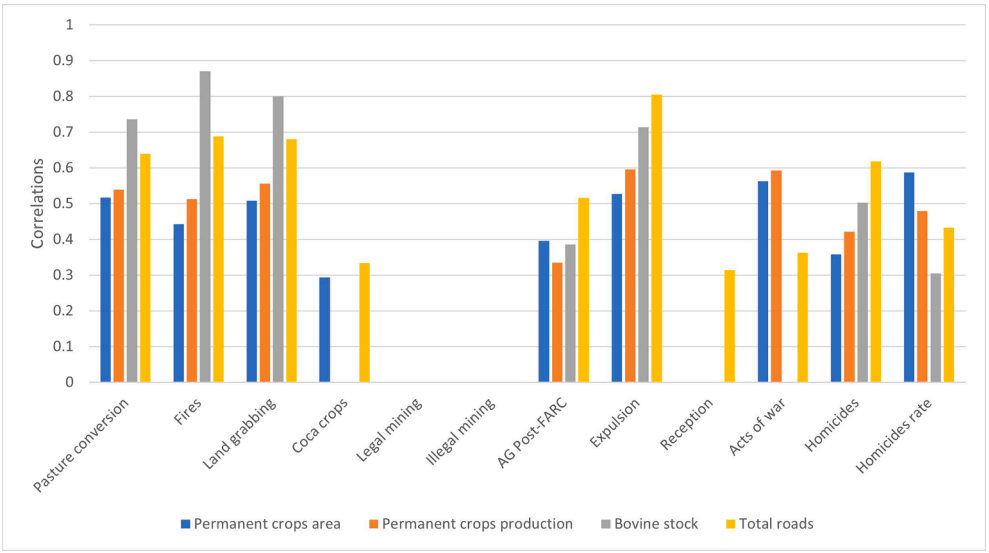

Fig. 23. Correlations between socio-environmental conflicts and social variables.

(2021) identify seven drivers of deforestation in Colombia: 1) pasture expansion, 2) infrastructure development, 3) the expansion of agricultural activities in restricted areas, 4) extensive livestock farming, 5) illicit crops, 6) illegal logging, and 7) illegal mining. Our results support this classification for the first five drivers; however, we lack data on illegal logging and do not find a strong relationship with mining activity.

In terms of socioeconomic indicators, municipalities with greater forest cover tend to have smaller total populations, lower migration, higher poverty, and more limited connectivity—both in terms of roads and internet access. In addition, municipalities with a higher share of Indigenous population tend to exhibit greater levels of natural forest cover.

For deforestation and forest fragmentation, the relationship with socioeconomic indicators is reversed. Municipalities with higher levels of deforestation and forest fragmentation tend to have larger migrant populations, greater areas under permanent crops, and more extensive road networks. As expected, municipalities with higher shares of Indigenous population tend to exhibit lower levels of deforestation and fragmentation.

The Supplementary Material outlines the methodologies used to explore relationships that may fall outside the scope of the main approach.

4.3. *Econometric analysis*

Our empirical approach combines three complementary components: (i) baseline correlations, (ii) 2SLS estimates to address potential endogeneity, and (iii) machine learning models to assess predictive relevance. The objective is to identify robust patterns of association between deforestation and a set of socioeconomic, conflict-related, and biophysical indicators.

4.3.1. *Structural model*

Table 3 reports OLS estimates of the structural determinants of deforestation. In the baseline specification (Column 1), the presence of post-FARC armed groups is positively and strongly associated with deforestation. This association remains positive, although statistically insignificant, when only structural controls are included (Column 4). However, once the socio-environmental conflict variables are incorporated (Columns 3 and 5), the estimated coefficient becomes negative and statistically insignificant. This pattern suggests that the positive association between post-FARC armed-group presence and deforestation is largely mediated by socio-environmental conflict channels, particularly land grabbing, rather than reflecting a direct relationship with forest loss.

Land grabbing—proxied by pasture expansion—emerges as the most

Table 3
Structural determinants of deforestation.

	Dep. Variable: Deforestation (ha) in 2019				
	(1)	**(2)**	**(3)**	**(4)**	**(5)**
Post-FARC armed groups	1,629***		−535.2	101.6	−577.9
	(578.8)		(444.9)	(574.9)	(492.4)
Land grabbing		0.24***	0.26***		0.26***
		(0.03)	(0.03)		(0.61)
Coca cultivation		0.11**	0.16**		0.19
		(0.044)	(0.064)		(0.12)
Illegal mining		2.30	2.44		6.88
		(2.84)	(3.77)		(5.67)
Migration (%)				2,157	−1,286
				(1,335)	(1,057)
Forest cover (%)				4,988***	266.4
				(1,858)	(644.3)
Fragmented Forest (%)				−10,953	−1,886
				(14,622)	(10,151)
Indigenous population (%)				−1,291	−30.97
				(1,780)	(839.1)
Poverty index				26.59	15.61
				(21.91)	(10.08)
Type 1 Roads				−32.57	6.29
				(20.23)	(5.69)
Type 3 Roads				−21.48	21.17
				(78.51)	(37.82)
Type 5–7 Roads				4.70***	−0.05
				(0.90)	(0.88)
Zone				1,770	1,120
				(1,272)	(690.5)
Constant	797.2**	64.35	138.4	−6,013**	−1,192
	(318.5)	(81.89)	(71.77)	(2,409)	(880.7)
Zone FE	No	No	No	Yes	Yes
Observations	77	77	77	77	77
R-squared	0.201	0.852	0.866	0.599	0.874

Notes: The dependent variable is deforestation (hectares) in 2019. All models are estimated using OLS with standard errors clustered at the municipality level, reported in parentheses. Column (1) includes only the post-FARC armed groups variable. Column (2) includes only the socio-environmental conflict variables (land grabbing, coca-cultivation, and illegal mining). Column (3) combines the post-FARC armed groups variable with the socio-environmental conflict variables. Column (4) reports the baseline specification including all structural controls but excluding land grabbing, coca cultivation, and illegal mining. Column (5) presents the full specification, combining the structural controls included in Column (4) with the socio-environmental conflict variables. Zone fixed effects are included in both Columns (4) and (5) to account for regional heterogeneity between the northwestern (zone = 1) and eastern (zone = 0) regions of the Colombian Amazon Basin. Land grabbing is proxied by pasture expansion, while coca cultivation and illegal mining capture alternative land-use pressures. The presence of post-FARC armed groups proxies for post-conflict dynamics and it is constructed as an intensity measure ranging from 0 to 3, where 0 indicates municipalities with no armed group presence during the 2017–2019 period, and 3 indicates municipalities where armed groups were present in all years analyzed. Forest area and forest fragmentation measure baseline ecosystem conditions. Migration, poverty, Indigenous population, and road types capture demographic and socioeconomic characteristics. Statistical significance is denoted as follows: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

robust correlate across specifications. It is positively and statistically significant both in the conflict-only specification (Column 2), when combined with post-FARC armed groups (Column 3), and in the full specification (Column 5). By contrast, coca cultivation is positively associated with deforestation in simpler specifications (Columns 2 and 3) but loses statistical significance once the full set of controls is included (Column 5), indicating that its partial association with deforestation is not robust to the inclusion of additional covariates. Meanwhile, illegal mining does not exhibit a statistically significant association with deforestation in any specification.

The baseline specification (Column 4), which includes the full set of structural controls but excludes the socio-environmental conflict variables, shows that small roads are positively and significantly associated with deforestation. However, once the socio-environmental conflict variables are incorporated in the full specification (Column 5), the variable is not statistically significant.

Finally, comparing the baseline specification (Column 4) with the full specification (Column 5) indicates that land grabbing plays a relevant role, but the model suggests that endogeneity is an issue in the estimations.

4.3.2. *Two-stage model estimates*

As outlined in the Methods section, the first step involves estimating a regression for each socio-environmental conflict, based on variables identified in the literature and the findings from the correlation analyses. Table 4 reports the first-stage regressions for each endogenous variable. The results reveal heterogeneous patterns of association between the instruments and endogenous processes.

For land grabbing (Column 1), the presence of post-FARC armed groups is positive and statistically significant, suggesting a strong association between post-conflict dynamics and land appropriation. Land grabbing is more closely associated with type-5 to type-7 roads. Migration also exhibits a positive and statistically significant relationship. By contrast and as expected, forest fragmentation is negatively associated with land grabbing, as it represents an earlier stage in the deforestation process. Other variables—including Indigenous

Table 4
First-stage results for socio-environmental conflicts.

Variables	**Land grabbing (1)**	**Coca Cultivation (2)**	**Illegal Mining (3)**
Post-FARC armed groups	4,877***	−13.29	0.30
	(1,595)	(278.6)	(0.72)
Type 1 roads		−11.08**	
		(4.42)	
Type 3 roads		187.1***	
		(62.95)	
Type 5–7 roads	12.09***		
	(2.90)		
Type 7 roads			0.004
			(0.005)
Forest Fragmentation (%)	−101,722*		
	(53,468)		
Forest cover share			2.65
			(2.22)
Indigenous population (%)	−1,095	−935.9**	−6.20**
	(4,136)	(389.4)	(2.66)
Poverty index	77.29	14.37	−0.10
	(71.49)	(9.14)	(0.09)
Migration (%)	11,339**		
	(5,312)		
Zone	880.9	714.4*	−14.49*
	(3,533)	(418.8)	(7.59)
Constant	−7,600	−477.1	18.58
	(6,333)	(879.5)	(11.86)
Observations	77	77	77
R^2	0.614	0.263	0.197

Notes: This table reports first-stage regressions for the endogenous land-use variables: land grabbing (Column 1), coca cultivation (Column 2), and illegal mining (Column 3). All models are estimated using OLS with standard errors clustered at the municipality level, reported in parentheses. The set of regressors varies across specifications depending on the outcome variable. Post-FARC armed groups proxy for post-conflict dynamics and it is constructed as an intensity measure ranging from 0 to 3, where 0 indicates municipalities with no armed group presence during the study period, and 3 indicates municipalities where armed groups were present in all years analyzed. Road types capture differences in transport infrastructure, while migration reflects population pressures. Forest fragmentation and forest cover share characterize baseline ecosystem conditions. Indigenous population and the multidimensional poverty index capture demographic and socioeconomic conditions. Zone is included as a control for spatial heterogeneity between the northwestern and eastern regions of the Colombian Amazon Basin. Statistical significance is denoted as follows: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

population, poverty, and zone—are not statistically significant in explaining land grabbing.

For coca cultivation (Column 2), the presence of post-FARC armed groups is not statistically associated with coca cultivation once the remaining covariates are included. This lack of association can be explained by the shift in coca cultivation in Colombia from the Amazon basin toward the departments of Norte de Santander and Nariño, driven in part by their proximity to key drug trafficking routes. In the CAB, coca cultivation is negatively associated with primary roads (Type 1) and positively associated with secondary roads (Type 3), suggesting that coca activity tends to concentrate in areas that are accessible but relatively remote from major transportation networks and urban centers. Indigenous population is negatively related to coca cultivation, whereas the zone indicator is positive and marginally significant.

For illegal mining (Column 3), the first-stage results show weaker explanatory power and limited statistical significance. Neither post-FARC armed groups, forest cover, road infrastructure, nor poverty are statistically significant predictors of this activity. By contrast, Indigenous population and the zone indicator are both negatively and significantly associated with it. These results suggest that illegal mining tends to occur in relatively isolated areas with a higher indigenous presence, particularly in the eastern Colombian Amazon, a region characterized by extensive natural forest cover.

The first stage results suggest that armed conflict—measured by the presence of post-FARC armed groups— is strongly associated with land grabbing but not with coca cultivation or illegal mining. This pattern suggests that post-conflict dynamics are more closely linked to processes of land appropriation than to other illicit land-use activities during the study period. One possible explanation is that, during the study period, coca cultivation and illegal mining were influenced by a broader set of economic, geographic, and institutional factors, which may not be adequately captured by contemporaneous measures of armed-group presence.

The key results for the analysis are those from the second stage, presented in Table 5. The most important finding is that, within our empirical framework, land grabbing emerges as the socio-environmental conflict most strongly associated with deforestation. By contrast, neither coca cultivation nor illegal mining is statistically associated with deforestation, consistent with our previous results.

Among the control variables, the poverty index is positively associated with deforestation in the coca cultivation and illegal mining specifications and is statistically significant in both. The Indigenous population share is negatively and significantly associated with deforestation in the coca cultivation specification—consistent with findings in the literature emphasizing their role in forest conservation. Finally, the zone indicator is positive and statistically significant in coca cultivation specification, suggesting that regional differences remain relevant even after accounting for the instrumented socio-environmental conflict.

Identification diagnostics indicate that the land grabbing model is identified and the excluded instruments are reasonably relevant, although the Kleibergen–Paap statistic falls slightly below the corresponding Stock–Yogo critical value. By contrast, coca cultivation and especially illegal mining specifications exhibit evidence of weak identification. The Hansen J test fails to reject the null hypothesis of instrument validity in all three specifications, providing no statistical evidence against the validity of the overidentifying restrictions.

4.3.3. Machine learning evidence

The XGBoost results (Table S-6 of the supplementary material) provide a complementary assessment of variable relevance within a flexible, non-parametric framework. In the full specification, land grabbing accounts for the largest share of predictive gain, indicating that it contributes most to reducing model error, followed by burned areas, which captures fire-based land clearing. Other variables—including cattle stock, degraded forest, and coca cultivation—make comparatively smaller contributions, while forest fragmentation, despite its broad coverage across observations, has a limited marginal impact on predictive performance.

When the model is restricted to socioeconomic and conflict-related variables, land grabbing remains the dominant predictor by a wide margin, with all other variables contributing far less to model fit. In this specification, burned areas, coca cultivation, and cattle stock retain some predictive relevance, whereas conflict-related indicators (e.g., homicides, post-FARC armed groups) and illegal mining make negligible contributions across all important metrics. See Section 3 of the Supplementary Material for further details.

These results indicate that variation in deforestation is primarily associated with land appropriation and clearing processes, while the marginal contribution of conflict-related variables becomes limited once these proximate land-use dynamics are accounted for. This pattern is consistent with econometric evidence and supports an interpretation in which conflict-related factors operate indirectly—through their association with socio-environmental processes—rather than as direct predictors of deforestation.

Table 5
Results for the second stage: explaining the deforestation model.

	***Dep. Variable**: Deforestation (ha) in 2019*		
	(1)	**(2)**	**(3)**
Endogenous variable	Land Grabbing	Coca cultivation	Illegal mining
Endogenous variable	0.22***	0.61	2,300
	(0.02)	(0.64)	(2,701)
Poverty index	19.63	79.46***	298.5*
	(15.16)	(28.39)	(173.2)
Indigenous population (%)	−641.5	−2,683*	11,846
	(990.4)	(1,506)	(15,499)
Forest Fragmentation (%)	2,469		
	(11,659)		
Migration (%)	−1,523*		
	(858.5)		
Forest cover share			−2,558
			(9,787)
Zone	946.4	2,497*	35,851
	(639.4)	(1,275)	(26,848)
Constant	−736.6	−4,267**	−48,915
	(721.5)	(1,975)	(33,177)
Under identification (Kleibergen-Paap LM)	7.17 (p = 0.028)	3.68 (p = 0.298)	0.826 (p = 0.6616)
Weak identification			
Cragg–Donald F	28.84	3.72	0.22
Kleibergen–Paap F	16.64	6.15	0.35
Stock–Yogo 10% threshold	19.93	22.30	19.93
Overidentification (Hansen J)	2.79 (p = 0.095)	3.87 (p = 0.1444)	0.00 (p = 0.9889)
Observations	77	77	77
# Excluded instruments	2	3	2

Notes: All models are estimated using OLS with standard errors clustered at the municipality level, reported in parentheses, where the endogenous variable indicated in each column is instrumented using the corresponding set of excluded instruments from the first stage. The Kleibergen–Paap LM statistic tests for under identification (null: the model is under identified). Weak identification is assessed using the Cragg–Donald and Kleibergen–Paap F-statistics; Stock–Yogo critical values correspond to the 10% maximal IV relative bias threshold. Overidentification is evaluated using the Hansen J test (null: instruments are jointly valid). All specifications include the control variables reported in the table. The number of excluded instruments is reported for each specification. Statistical significance is denoted as follows: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

5. Discussion

This study pursued two main objectives. First, it characterizes the Amazon biome using social and environmental indicators. Second, it estimates explanatory models of deforestation in the Colombian Amazon biome, drawing on indicators across multiple categories. The characterization of the CAB was based on four analytical dimensions: social (socioeconomic and demographic information), forest cover, freshwater, and socio-environmental conflicts. Each dimension was organized into categories, which were then represented by a set of indicators. To address the second objective, we estimated Pearson correlations between indicators and categories across dimensions, applied machine learning tools to assess the strength of these relationships, and drew on the literature to identify relevant associations. We then estimated a two-stage model to explain deforestation across the Colombian Amazon biome.

The analysis of socioeconomic factors in the CAB is constrained by limited data availability compared to other parts of the country. Just as access to many territories in the Amazon is difficult, the collection of reliable data to understand the social and economic dynamics of its population is equally challenging. This limitation hinders the monitoring of economic, demographic, and productive activities, as well as a range of socio-environmental conflicts.

Despite these challenges, the available data provide a valuable basis for understanding the socioeconomic conditions of the population in the Colombian Amazon—particularly spatially explicit information, which is increasingly available in greater quantity and quality, often in near real-time, even for remote areas. Such data enables detailed detection of land cover changes, including pasture expansion and deforestation.

The municipal-level analysis further highlights the substantial heterogeneity among CAB municipalities, reflecting geographic, biophysical, and historical differences across all dimensions. This information not only provides a regional (CAB) perspective but also helps to capture the specific conditions of smaller administrative units, supporting the design of policies, plans, and strategies that align national objectives with local needs.

Using available spatial and socioeconomic data, numerous studies have examined deforestation in Colombia to identify its drivers. However, most do not reach a consensus or provide definitive conclusions. Given the complexity, variability, simultaneity, dynamism, and interdependence of deforestation drivers, we emphasize the value of correlation-based frameworks as a first step before attempting causal inference through econometric models. The inherent complexity of deforestation in the Amazon, combined with limited data availability, makes it challenging to identify and quantify its main drivers and their relationships with forest loss. In this study, we address these challenges by constructing a comprehensive set of indicators that capture the region's economic, social, and environmental dynamics. We then estimate the relationship between key drivers and deforestation using econometric models and apply machine learning techniques to uncover additional patterns and associations within the high-dimensional set of variables.

Among the key findings, we highlight higher levels of poverty in the biome's population compared to the national average, particularly in municipalities with larger shares of Indigenous populations. We also found significant variation across municipalities in forest cover, deforestation, illicit crop cultivation, armed conflict, pasture expansion, fires, and illegal mining. More importantly for this study, there were marked socioeconomic differences between the departments in the foothill zone (Northwestern Amazon) and those in the plain's region (Eastern Amazon). The northwestern region has a stronger economy and lower poverty levels. However, it is also a region where socio-environmental conflicts (except for illegal mining) are more intense or frequent. This region also has greater road density, connectivity, agricultural production, and livestock expansion, suggesting that development and deforestation pressures are closely intertwined.

Consistent with the literature, our results identify land grabbing as the most relevant socio-environmental conflict associated with deforestation across specifications. By contrast, neither coca cultivation nor illegal mining exhibits a statistically significant association with deforestation once structural factors are considered (Busch & Ferretti-Gallon, 2023; Armenteras et al., 2013a, 2017; González-González et al., 2021a; Hänggli et al., 2023; Lim et al., 2017; Tebbutt et al., 2021). The estimated coefficients associated with Indigenous population are generally consistent with the literature, although statistically significant only in selected specifications, suggesting that Indigenous governance remains an important correlate of forest conservation.

The absence of robust evidence linking alluvial mining to deforestation in our estimates should be interpreted with caution. First, measurement errors are likely substantial, as illegal mining activities are systematically underreported in official datasets, particularly in remote and conflict-affected areas. This introduces attenuation bias, potentially biasing the estimated coefficients downward. Second, our results pertain to the 2018–2019 period, which coincides with relatively moderate international gold prices. By contrast, the subsequent period—particularly 2022–2026—has been characterized by a substantial increase in gold prices (Figure S-24), which is well documented to incentivize the spatial expansion and intensification of illegal mining activities (Asner et al., 2013). These considerations suggest that the lack of statistical significance should not be interpreted as evidence of absence of an effect. Rather, it reflects both data limitations and a temporally specific equilibrium in which mining pressures were comparatively lower. Moreover, illegal mining in Colombia is typically concentrated at specific riverine sites, with more severe impacts on water bodies than forest cover. In this sense, illegal mining remains a potentially important and growing driver of pollution and erosion, more than deforestation, warranting continued monitoring and policy attention as market conditions and the presence of armed groups evolve.

A similar phenomenon is observed for the case of coca cultivation. Changes in government policies toward post-FARC armed groups have been associated with notable increases in coca cultivation across Colombia, including the CAB, during the 2022–2026 period.

These findings help clarify several debates in literature. First, they support the view that armed conflict and territorial control have complex, context-dependent effects: areas under the influence of post-FARC armed groups may experience both forest protection (due to restricted land access) and deforestation (driven by land grabbing and, potentially, illicit crop expansion). Second, our results reinforce the literature showing that structural and biophysical conditions—such as road access—shape where and how socio-environmental conflicts translate into deforestation (Dávalos et al., 2016; Hänggli et al., 2023). Third, they underscore the interaction among poverty, Indigenous governance, land-use incentives, and conflict dynamics, suggesting that deforestation cannot be explained by single-factor models.

As noted above, much of the conservation in the CAB occurs in areas where local well-being is precarious, with communities facing extreme poverty and limited access to effective social services to meet basic needs such as health, education, and housing. The unequal distribution of biodiversity benefits remains a pressing issue in the CAB. In this context, the role of Indigenous communities in forest conservation underscores the need to strengthen support for community governance and to prioritize the enforcement of territorial rights and boundaries.

Our analysis points to several specific policy-relevant conclusions: i) Given the central role of land grabbing as the primary proximate driver of deforestation, policy efforts should prioritize strengthening land tenure security, improve cadastral systems, and enforce property rights in frontier regions. This includes targeting illegal land appropriation on public lands, forest reserves, and protected areas, as well as enhancing monitoring and sanctioning capacity to prevent the consolidation of illicit land claims. ii) Although coca cultivation was not statistically associated with deforestation during the study period once broader structural factors were taken into account, policies aimed at promoting

sustainable livelihoods and reducing dependence on illicit economies—such as alternative livelihood programs, sustainable agriculture incentives, and crop-substitution strategies— remain important complements to broader forest conservation efforts. iii) In regions affected by post-conflict dynamics, interventions should combine conflict-sensitive land management with environmental monitoring to prevent opportunistic deforestation by illegal actors. iv) Road construction and connectivity should be carefully planned, with environmental impact assessments and enforcement mechanisms to minimize unintended deforestation. Policies could prioritize areas with lower forest fragmentation and high conservation value, and v) Effective forest conservation requires coordination among state authorities, Indigenous communities, and non-state actors to manage land-use conflicts, monitor deforestation, and provide sustainable economic alternatives.

Understanding and monitoring the socioeconomic conditions of the CAB's inhabitants—living in a globally strategic conservation area—helps identify and implement nature- and community-based approaches that support the long-overdue well-being of these populations. We also hope that this characterization and analysis of interactions between indicators and categories will shed further light on the drivers of deforestation in the Amazon biome. In this way, our study advances understanding of deforestation in the Colombian Amazon by linking structural socioeconomic conditions, armed conflict, and land-use processes to observed forest loss. The evidence highlights the need for interventions that are spatially targeted, conflict-sensitive, and integrated across socioeconomic and ecological dimensions, in line with the broader literature advocating multi-level and context-specific strategies for forest conservation.

CRediT authorship contribution statement

Jorge Higinio Maldonado: Writing – review & editing, Writing – original draft, Validation, Supervision, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Rocío del Pilar Moreno-Sánchez:** Writing – review & editing, Writing – original draft, Supervision, Investigation, Formal analysis, Conceptualization. **Jorge Montero-Mestre:** Writing – review & editing, Validation, Software, Methodology, Investigation, Formal analysis, Data curation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.worlddev.2026.107574.

Data availability

Data will be made available on request.

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